When Does Relational Structure Help Healthcare Fraud Detection? A Controlled Benchmark of Heterogeneous Graph Neural Networks and Graph-Augmented Tabular Models

H M Akshay[1*], Gaurav S B[1], Eshita Narayana[1], Harshita Srivastava[1], Malini M Patil[1]

[1]Department of Computer Science and Engineering, RV Institute of Technology and Management, Bengaluru, 560083, Karnataka, India.

*Corresponding author(s). E-mail(s): hmakshay_23bcs106.rvitm@rvei.edu.in;
Contributing authors: gauravsb_23bcs258.rvitm@rvei.edu.in; eshitan_23bcs031.rvitm@rvei.edu.in; harshitas_23bcs323.rvitm@rvei.edu.in; malinimp.rvitm@rvei.edu.in;

Abstract

Purpose: Graph neural networks (GNNs) are proposed for health-insurance fraud because collusive provider rings leave traces in claim networks; when they beat tabular models is unclear. **Methods:** We built a synthetic claims benchmark that independently varies the feature-level signal, doctor–patient density, claim volume and shape of fraud rings, and tests without leakage on unseen ring members. Sixteen methods, including gradient boosting with and without engineered graph features, five heterogeneous and two fraud-specialised GNNs, were compared in nine regimes, at four ring volumes, for five ring shapes and on public provider-labelled claims. **Results:** At default volume (ring patients filing 5.7 times the average), per-entity aggregates detected most rings (average precision 0.89–0.92) and graph features added up to 0.10 for dense rings. At 4–8 claims per month, aggregates fell to 0.05–0.42, graph features reached 0.41–0.92 for denser rings, and pure GNNs trailed them by 0.17–0.57, also after a wider search. No pure GNN outperformed gradient boosting with graph features averaged over regimes (upper 95% limits $\leq$0.006) or for other ring shapes; hybrids gained 0.01, and GNNs were 0.09–0.23 worse on isolated fraud. Recurring ring members inflated performance by up to 0.60. On public data, gradient boosting on provider

aggregates was best (0.704), also for expert-flagged group fraud. **Conclusion:** Relational structure helps when collusion densely links otherwise unremarkable members, and engineered shared-patient features captured it better than the GNNs tested. Graph models should be compared with graph-augmented tabular baselines on unseen entities.

Keywords: Healthcare fraud detection, Provider rings, Heterogeneous graph neural networks, Gradient boosting, Synthetic benchmark, Data leakage

1 Introduction

Health insurance fraud causes losses estimated in the tens of billions of dollars each year in the United States alone [1]. Detecting it is a routine task of payers' claims-processing and audit systems, which must decide from administrative claims data which providers and claims a limited team of investigators examines. Fraud takes two broad forms. In *isolated* fraud a single provider inflates or fabricates individual claims, so the evidence lies in the claim itself: amounts, procedure counts or coding that are unusual for the provider's specialty. In *collusive* fraud a group of providers and recruited patients file claims together. Each claim can look ordinary; the evidence lies in who bills whom—patients who see many members of the same small group of providers, and providers who share an unusually large fraction of their patients.

The second form motivates graph neural networks (GNNs) over the patient–provider–facility network, in particular heterogeneous architectures such as RGCN [2], HAN [3], HGT [4] and Simple-HGN [5]; GNNs of this kind have also been built specifically for health-insurance claims [6–8]. Whether GNNs are needed, however, is an empirical question. Gradient-boosted trees remain strong on tabular data [9], relational signals can be engineered as features, and benchmark studies outside healthcare report that tree ensembles with neighbourhood features often match or beat GNNs on graph anomaly detection [10, 11]. On public Medicare claims with provider fraud labels, machine-learning models given graph-centrality features outperformed GNNs at a fraction of the training cost [12]. A single dataset of this kind, however, cannot show *when* relational structure helps or *why*: its labels mark providers rather than rings, and neither the typology nor the strength of the relational signal can be varied.

Answering the question for healthcare claims is difficult for two reasons. First, public claims data with labelled collusion are all but absent, and real labels conflate fraud typologies. The only such labels we found are group-fraud flags that experts added, by an undocumented criterion, to the most widely used public provider-labelled dataset [13], on which reported results vary widely between studies and are rarely accompanied by significance tests or code [14]. Second, evaluation is fragile: identifiers that encode the label, statistics computed on the full dataset, and providers that appear in both training and test data inflate apparent performance [15, 16]. Fraud rings make the last problem acute, because a model can score a test claim correctly by recognising a provider it saw labelled during training instead of recognising coordination.

We therefore study the question in a controlled setting and check it on public provider-labelled data. Our contributions are:

1. **A map of when relational structure helps.** We vary three properties of collusive rings independently—how anomalous their claims are (δ), how densely their doctors and patients are linked (s), and how many claims they file—check four further ring shapes and sizes, and compare 16 methods, 13 of them tuned with an equal budget of configurations: tabular models with and without engineered graph features, a multilayer perceptron (MLP) control, five heterogeneous GNNs, two GNNs designed for fraud detection, two hybrids and an unsupervised overlap score. The result is a map of the conditions under which structure adds information beyond per-entity features, and of whether message passing or a few engineered features capture it (Sections 6 and 9).
2. **A reusable, leakage-tested benchmark.** The generator's *emergent* protocol forms every test ring from doctors and patients never seen in a training ring, and its audit-period design keeps features and graphs free of future information; automated tests enforce both (Section 3). Code, generator and tests are publicly available.
3. **Evaluation pitfalls, quantified.** Test rings whose members recur from training inflate ring detection by up to 0.60 average precision and can reorder methods; and pooled tests over generated worlds that share their background data overstate significance, so the generator seed, not the world, must be the unit of replication. A re-analysis of a small fixed benchmark shows why such data cannot separate relational learning from claim-level anomaly detection or from recognising known members (Sections 7 and 9.3).
4. **Mechanism.** Degree-preserving rewiring, ablations and an analysis of which inputs the trained models use show that at default volume GNNs detect dense rings from counts of who bills whom. A pre-declared test shows that at low volume two shared-patient columns alone recover most of the GNNs' deficit, although gradient boosting does not depend on those two (Section 7).
5. **A check on public data**: a provider-labelled claims dataset, described by its publisher as Medicare data, evaluated with disjoint training, validation and test providers, equal tuning budgets, repeated cross-validation, per-rotation model selection and tests corrected for overlapping training sets (Section 5). It replicates, under stricter conditions, an earlier finding on public Medicare provider data [12]. Its labels identify providers, not rings, so it tests whether graph information helps provider-level detection, not whether rings are detected. Published expert flags for group fraud on the same data [13] let us repeat the check on providers marked as organised fraud (Section 8).

Our main findings are as follows. Claim features alone cannot reveal rings whose claims look legitimate. When ring members are unusually active, as in our default setting, per-entity aggregates capture most of the evidence; as ring volume falls, aggregates lose most of their power and coordination structure carries most of the evidence, until rings that file about one claim per member are found by no method. Structure adds most when the ring's doctor–patient block is moderately or highly dense. Where it does, gradient boosting with a few engineered shared-patient features was as good

as or better than every pure GNN we tested, including two designed for fraud detection and across five ring shapes and sizes, and far better when ring volume was low, also after a three-times wider hyperparameter search; GNNs that also receive those features gained a small amount in average precision (AP; +0.01 averaged over regimes, of the order of the variation between training runs), most for sparse rings, where predeclared tests on fresh seeds and over three training seeds confirmed it at low volume. Some pure GNNs beat that gradient-boosting model, but only for sparse rings at reduced volume whose claims were also anomalous, where structure added little and logistic regression without graph inputs beat it too. GNNs are clearly worse on isolated fraud, consistent with message passing diluting claim-level anomalies. At default volume HGT's ring detection is structural: it collapses without degree encodings and survives with degree encodings as its only inputs. Routing high-risk claims from gradient boosting to a GNN gained nothing; fusing the two models' rankings gained at most 0.006 AP, as an ensemble rather than as a router. Evaluations that let test rings share members with training rings overstate ring detection, by up to 0.60 AP for a model that only sees entity attributes. On the public provider-labelled data, where labels show little relational assortativity, no graph-based method improved on gradient boosting over provider aggregates, in line with earlier work on these data [12]. The same held for 100 providers that experts flagged as group fraud, who are unusually large, the condition under which our benchmark predicts that aggregates suffice.

2 Related Work

Healthcare fraud detection.

Most published healthcare-fraud detectors are supervised classifiers over claim- or provider-level features: classical learners and neural networks on Medicare data labelled with the federal exclusion list (LEIE) [17–19], and multidimensional outlier analysis on Medicaid data [20]. In these studies relational information enters, if at all, through hand-engineered features. On the public provider-labelled dataset used in Section 5, reported F1 scores range from 0.49 to 0.90 under differing protocols, and significance tests and released code are rare [14].

Graph-based fraud detection.

Coordination leaves structural traces such as dense blocks, lockstep behaviour and unusual egonets [21, 22], which unsupervised detectors exploit directly [23–25]. GNNs have been applied to review fraud [26], underground-forum key players [27] and camouflaged or class-imbalanced fraudsters [28, 29], and spectral analyses argue that standard message passing can blur anomalies [30]; we include CARE-GNN and the spectral BWGNN among the compared methods. Heterogeneous GNNs use relation-specific weights (RGCN [2]), attention over hand-chosen meta-paths (HAN [3]) or attention parameterised by node and edge type (HGT [4]); Simple-HGN [5] showed that a GAT with edge-type embeddings is competitive with them when baselines are tuned fairly.

Graph learning for health-insurance fraud.

On data from single payers, heterogeneous GNNs have classified patients with meta-path attention [6] or graph structure learning [7], and billed activities with several heterogeneous GNNs and post-hoc explainers [8]; the baselines were other graph models [6] or tabular models without relational features [8]. On the public provider-labelled dataset, a graph transformer network reached an F1 score of 0.53 against 0.52 for a random forest in a study of graph attention networks [14, 31], meta-path embeddings computed on a 10% claim sample matched tree ensembles in ROC-AUC (0.87–0.88) [32], and machine-learning models given graph-centrality features outperformed GNNs in precision, recall and F1 while training 250–300 times faster [12]; our analysis of these data replicates the last finding under a stricter protocol. Other work targets organised fraud directly: a parallel GNN combines heterogeneous attention on the beneficiary–provider graph with community-aware risk diffusion to find medical-insurance fraud gangs, on the public dataset above with group-fraud labels added by experts [13], community detection on a co-visit network supports auditors looking for collusive groups [33], and graph analysis of patient–doctor–pharmacy networks has been deployed against fraud, waste and abuse [34]. These studies target organised groups directly. Their data are fixed, however, so neither the typology nor the strength of the relational signal can be varied, and they cannot show under which conditions relational structure carries information that per-entity features miss.

When do graph models beat tabular models?

Evidence is mixed. A random forest on node features outperformed graph convolutional networks on the Elliptic Bitcoin data [10]; tree ensembles with simple neighbourhood aggregation were among the strongest supervised graph anomaly detectors across ten datasets [11]; gradient-boosted trees with graph-based input features were very strong, though not universally winning, baselines on industrial graphs [35]; a graph gradient-boosting machine was on par with or better than the state of the art for insurance fraud [36]; and tree ensembles remain hard to beat on tabular data [9]. Theory offers one possible reason: message-passing networks cannot count induced substructures of three or more nodes [37], and coordination is a substructure. Synthetic benchmarks with injected typologies make the question testable, as for money laundering [38] and tabular fraud [39]. We found no comparable controlled study for heterogeneous patient–provider–facility claim graphs, where the relevant typology is the collusive provider ring.

Evaluation pitfalls.

Features that encode the target, preprocessing fitted on the full dataset and entities shared between training and test data inflate reported performance [15, 16]. GNN benchmarks are also sensitive to split choice and tuning effort [40], and duplicate nodes that leak between training and test data strongly affected GNN performance on two popular benchmarks [41]. Fraud rings make entity overlap acute: a model can score a test claim correctly by recognising a provider it saw labelled in training rather than by recognising coordination.

3 Benchmark Design

3.1 Synthetic claims world

We simulate 6,000 patients, 300 doctors and 40 hospitals in five regions over four audit periods of six months (Fig. 1; Table A1). Each doctor has one of eight specialties, which sets the distribution of claim amounts, inpatient probability and procedure counts. Patients have heterogeneous utilisation (gamma-distributed claim rates with mean three claims per two years) and one to three regular doctors in their region; 85% of legitimate claims go to a regular doctor and the rest to a popularity-weighted random doctor. Claim amounts are log-normal given the specialty, procedure count and inpatient status.

Isolated fraud.

Three per cent of claims are upcoded claims filed through a patient's own regular doctor. Their log-amount, procedure count, inpatient probability and procedure-code level are shifted by two standard deviations; their structure is that of an ordinary claim.

Provider rings.

A ring consists of four colluding doctors and twelve recruited patients, drawn preferentially from high utilisers. During its active period the ring files 16 claims per month between linked doctor–patient pairs, clustered around monthly burst days. Two parameters are varied independently:

- the *feature shift* $\delta \in \{0, 1, 2\}$ applied to ring claims, in standard deviations of log-amount (with matching shifts in procedures, inpatient probability and coding level). At $\delta = 0$ a ring claim is drawn from exactly the distribution of a legitimate claim of the same doctor's specialty;
- the *structural density* $s \in \{0.1, 0.5, 0.9\}$, the probability that a ring patient is linked to each ring doctor. Ring claim volume is fixed, so s changes only how the claims are spread over the doctor–patient block (realised block densities ≈ 0.37, 0.62 and 0.79).

Ring patients keep a single regular doctor, so the number of distinct doctors they see is governed by s. Their elevated claim volume, a degree signal visible to every model, is the same in all regimes and large: in a test period a ring patient files on average 9.3 claims, against 1.6 for other patients with a claim (99th percentile about 5), and a ring doctor 38 claims against 16. Section 6.6 reduces ring volume to four claims per month, which brings ring patients to 3.7 claims, about the 95th percentile of other patients, and ring doctors to 20, and Section 6.7 replaces the block by other shapes and sizes.

Protocols.

Under the *emergent* protocol (default), four new rings start in each period and rings in different periods are formed by disjoint doctors and patients, so every test ring is unseen during training. Under the *persistent* protocol the same four rings are active in

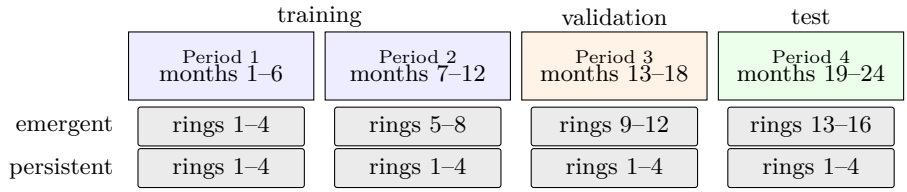


Fig. 1 Audit-period protocol. Each six-month period is analysed separately: features and graphs use only that period's claims. Under the emergent protocol every period has new rings formed by disjoint doctors and patients, so test rings are unseen; under the persistent protocol the same rings recur and can be recognised by their members.

all periods, which reproduces the setting of most prior synthetic evaluations and lets a model succeed by recognising known members. Both protocols have the same expected claim volume and the same ring-claim volume per period, but they are generated separately, so a persistent world does not share the background of the emergent world of the same seed. Ground truth (fraud type and ring identifier) is stored separately from the claim records and never enters feature construction.

Replication.

Each regime is generated from twelve seeds. The generator draws a world's entities, legitimate claims, isolated fraud and recruited ring members before it applies the ring parameters, so for a given seed the nine regimes share all of these and differ only in their ring claims. Regimes are therefore compared on otherwise identical worlds, and analyses pooled over regimes treat the seed, not the regime–seed world, as the unit of replication (Section 4.5).

3.2 Evaluation protocol

Periods 1–2 are used for training, period 3 for validation (early stopping, threshold selection) and period 4 for testing. Each period is analysed on its own, as in a retrospective audit: a claim's aggregate and graph features, and the graph a GNN operates on, are built only from claims of the same period (one label-free exception: the specialty means behind the claim's relative log-amount come from the training periods). This keeps every split's features on the same time scale and prevents information from later periods reaching earlier ones; it is verified by automated tests (Section 10).

4 Methods

4.1 Tabular features

- **C** (claim): log-amount, procedures, length of stay, inpatient flag, coding level, month (cyclical), patient age, sex and chronic-condition count, doctor experience and specialty, hospital size.
- **A** (aggregates within the period, no labels): the patient's claim count, distinct doctors and mean/maximum log-amount; the doctor's claim count, distinct patients and mean log-amount; hospital claim count; patient–doctor pair count; and the claim's log-amount relative to the patient, doctor and specialty means.

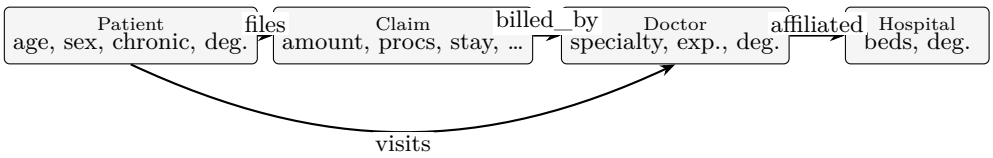


Fig. 2 Heterogeneous claim graph of one audit period. Every relation also has a reverse relation (not drawn), so every node type receives messages and doctor context reaches claim nodes. Entity nodes carry static attributes and log(1 + degree) per incident relation; claim nodes carry only claim-intrinsic attributes.

- **G** (graph features on the period's patient–doctor bipartite graph, no labels): the doctor's maximum shared-patient Jaccard similarity with another doctor and number of doctors sharing at least two patients, the mean degree of the doctor's patients, the patient's mean doctor overlap and share of doctor pairs that co-treat other patients, and the triadic closure of the claim (how many of the patient's other doctors share patients with the claim's doctor), raw and degree-normalised.

4.2 Heterogeneous graph

Each period's graph has patient, doctor, hospital and claim nodes and the relations *files* (patient→claim), *billed_by* (claim→doctor), *affiliated* (doctor→hospital) and *visits* (patient→doctor), each with a reverse relation so that every node type receives messages. Claim nodes carry claim-intrinsic attributes; entity nodes carry their static attributes and log(1 + degree) for each incident relation, a label-free structural encoding included because attention- and mean-based aggregators cannot otherwise count neighbours [42]. No engineered aggregates or labels are used as node features (Fig. 2). Claim attributes and degree encodings are standardised with statistics of the training periods; static entity attributes, which do not change over time and carry no labels, with statistics over all entities.

4.3 Models

- **Tabular**: logistic regression (LR) and random forest (RF) on C+A; histogram gradient boosting (GB) [43] on C, C+A and C+A+G. Scikit-learn's implementation stops early on a random 10% of the training claims when a world has more than 10,000 of them, as some seeds do; it never sees validation or test claims.
- **MLP-C**: a multilayer perceptron on C, the no-graph control for the GNNs.
- **GNNs**: heterogeneous GraphSAGE [44] (one convolution per relation, mean or sum aggregation), RGCN [2], HAN [3] (claim nodes; meta-paths claim–patient–claim and claim–doctor–claim, with patient, doctor and hospital attributes appended to claim inputs), Simple-HGN [5] and HGT [4]. All share the same per-type input projection, residual connections with layer normalisation, and two-layer classification head.
- **Fraud-specialised GNNs**: BWGNN [30], which replaces message passing by band-pass Beta-wavelet filters on each relation, and CARE-GNN [28], which keeps, on each relation, only the neighbours whose predicted labels are closest to the node's own, with thresholds adapted during training and an auxiliary loss that trains the

similarity measure. Both run on the claim-only relation graphs that HAN uses, keep their published layer structure and share the input projection and classification head; CARE-GNN is trained full-batch with the class-weighted loss used for every other model rather than with under-sampling.

- **Hybrids**: SAGE+AG and HGT+AG, whose claim nodes also receive the A and G features, testing whether message passing adds information beyond the engineered features. Hybrids reuse the tuned configuration of their base architecture and receive no separate tuning budget.
- **Overlap score**: the degree-normalised triadic closure of the claim, used directly as an unsupervised score.

Tabular models use scikit-learn [45] and graph models PyTorch Geometric [46]. Neural models are trained full-batch with AdamW, class-weighted binary cross-entropy and early stopping on validation average precision over all fraud (patience 20; at most 200 epochs, 300 for the MLP; GNN gradients are clipped to norm 1). Every model is trained once per world with the same training seed. Because a network's initial weights depend only on the seed and the layer sizes, every world, including the development worlds used for tuning, starts from the same initial weights, so variation between worlds does not include variation due to initialisation; a replication with two further training seeds is reported in Online Resource 1, Section S6.

Hyperparameters.

Every method receives the same budget of up to ten configurations (the first being a default), selected by mean validation average precision on two development worlds (generator seeds 901 and 902, $\delta = 0.5$, $s = 0.5$) that are never used for evaluation (Online Resource 1, Table S1). The development regime ($\delta = 0.5$) is deliberately not one of the evaluation regimes. Candidates are drawn once per method with a fixed sampling seed. An equal budget is not an equal share of the search space: ten configurations cover 7% of a GNN's grid of 144, 9% of gradient boosting's 108 and all six of logistic regression's, and every re-tuning reported below re-selected among the same candidates. Section 6.8 therefore reports a wider search for the strongest GNNs.

4.4 Two-stage routing

A common deployment proposal is to score every claim with a cheap tabular model and pass only high-risk claims to a GNN. We evaluate this with GB-CAG as the first stage and HGT or SAGE as the second. Claims whose GB-CAG score exceeds a threshold set on validation data are escalated, with the share of escalated claims chosen by validation AP, and are ordered either by the GNN score alone (*re-rank*) or by the mean of the two models' validation percentiles (*fused*). The stages are the models of the main experiment, retrained with the same seeds; their test predictions reproduce the stored ones exactly for the GNNs and to within 3×10^{-8} for GB-CAG (float32 storage). Details are in Online Resource 1, Section S2.

4.5 Metrics and statistics

The primary metric is average precision (AP, the area under the precision–recall curve), appropriate under class imbalance [47]. *Ring AP* scores ring claims against legitimate claims (isolated fraud excluded); *isolated AP* is defined analogously. We also report F1 at the threshold that maximises validation F1, the area under the receiver operating characteristic curve (ROC-AUC) and, because investigators can examine only a limited number of cases [48], recall when the top 5% of claims are investigated. For routing we add precision among the top 5% of claims (P@5%) and, as a ring-level measure, the share of test rings with at least one claim in that budget.

Each of the nine regimes is replicated with twelve worlds (generator seeds 1–12), 108 worlds in total. Seeds 1–6 formed a first round; seeds 7–12 were added in a second round whose size, analyses and reporting were fixed in a plan committed to the repository before any of its runs started, after the first round's results had been seen. A third round, planned in the same way, added three robustness checks: a feature shift at reduced ring volume, replication over training seeds and per-regime tuning (Online Resource 1, Sections S5–S7), a fourth added two fraud-specialised GNNs and four further ring shapes and sizes (Section 6.7), and a fifth widened the hyperparameter search (Section 6.8), tested the closure columns (Section 7.2) and repeated the confirmatory test with two more training seeds (Section 6.6). A sixth, planned in the same way after we found published group-fraud flags for the public data, re-scored that analysis on them (Section 8). Within a regime the twelve worlds are independent, and methods are compared with paired t-tests over them. Because the nine regimes of a seed share most of their data (Section 3.1), comparisons pooled over regimes average each seed's nine paired differences and test the twelve seed means (t-test with eleven degrees of freedom); treating the 108 worlds as independent would overstate the evidence. The persistent-ring, ablation, routing and ring-volume analyses use seeds 1–6. p-values are Holm-adjusted within each family of comparisons [49] (families were fixed in advance for rounds two to six, not for the first round), Wilcoxon signed-rank tests are a robustness check [50], and intervals are 95% confidence intervals (CIs) from the t-distribution over worlds or, for pooled comparisons, over seeds. A non-significant difference is not evidence that two methods are equivalent; the CIs show which differences remain compatible with the data.

4.6 Use of AI tools

An AI coding assistant (Claude, Anthropic) was used to help design the experiments, write and review the analysis code, run and document the experiments, and draft and edit parts of the text. All reported numbers are produced by the released code (Section 10). The authors reviewed all code, analyses and text and take full responsibility for the content.

5 Public Provider-Labelled Data

5.1 Data

We use the public "Healthcare Provider Fraud Detection Analysis" dataset [51] (CC0 public domain), which its publisher describes as Medicare inpatient and outpatient claims linked to beneficiaries, with each provider labelled as potentially fraudulent or not; graph models studied on these data are discussed in Section 2. Only the labelled training files are used: 558,211 claims (40,474 inpatient, 517,737 outpatient) dated between November 2008 and December 2009, filed by 5,410 providers (506, or 9.4%, labelled fraudulent) for 138,556 beneficiaries, with 100,737 distinct attending, operating or other physicians. The documentation describes neither the source of the claims nor how the labels were obtained, so we treat them as given and discuss this limitation in Section 9.6. The beneficiary file resembles the CMS synthetic public-use files (all 138,556 dates of birth fall on the first of a month and chronic conditions are coded 1 and 2), so the records may be partly or wholly synthetic; we could not establish this. Because labels exist only per provider, the prediction unit is the provider; claim-level ring detection cannot be evaluated on these data.

Group-fraud flags.

Ma et al. [13] added a second label to the same claims: anti-fraud experts flagged 100 of the 506 fraudulent providers as involved in organised fraud, and the flags are public. The criterion is not documented, and neither group identifiers nor group sizes are published. The release recodes identifiers; we mapped them to the original ones by rank and confirmed the mapping for all 5,410 providers on claim counts, amounts and fraud labels.

5.2 Features and graph

P (23 provider aggregates): claim volume and inpatient share, beneficiaries and claims per beneficiary, distinct attending and operating physicians, reimbursement totals and log-amount statistics, deductibles, claim duration and length of stay, diagnosis and procedure counts, share of claims with an operating physician, and the age, chronic-condition burden, mortality, renal-disease share, annual reimbursements and geographic spread of the provider's beneficiaries. **G** (8 label-free graph features): for both shared beneficiaries and shared physicians, the number of co-providers, the maximum Jaccard overlap with another provider, the share of the provider's beneficiaries (physicians) seen by other providers, and their mean number of providers. **L** (4 label-neighbourhood features): the share of co-providers that are known fraudulent *training* providers, and an indicator of having any labelled co-provider, through beneficiaries and through physicians. L uses only training labels and never a provider's own label.

The heterogeneous graph has provider, beneficiary and physician nodes, with relations *serves* (provider–beneficiary, 363,300 edges) and *employs* (provider–physician) plus reverse relations. Physicians working for a single provider carry no cross-provider information and are summarised in P rather than kept as nodes, leaving 7,685 shared

physicians (16,287 edges). Provider nodes carry P (or P+G for the hybrids), beneficiary nodes 22 demographic and clinical attributes, and physician nodes their claim and provider counts. HAN, BWGNN and CARE-GNN use the provider-only meta-paths provider–beneficiary–provider and provider–physician–provider.

5.3 Protocol

All providers' claims are observed and define the graph; labels of validation and test providers are hidden. This matches an audit in which claims are available for every provider but only some providers have been investigated. Providers are assigned to five stratified folds; in rotation k, fold k is the test set, fold $k+1$ the validation set and the remaining three folds the training set, so every provider is tested exactly once. Hyperparameters are selected once, with the same budget of six configurations per method, on a separate stratified 60/20/20 split of the same providers (seed 1000), and then fixed for the five rotations. Graph models are trained full-batch for at most 100 epochs with early stopping (patience 10) on validation AP. Metrics are AP, ROC-AUC, F1 at the validation-selected threshold, and recall among the top 10% of providers by score (the label prevalence is 9.4%). Because the rotations' training sets overlap, their results are positively correlated and a plain paired t-test over them is anti-conservative. Differences between methods are therefore tested with the corrected resampled t-test [52], which replaces the factor $1/J$ in the variance of the mean difference by $1/J + n_2/n_1$, where $J = 5$ is the number of rotations and $n_2/n_1 = 1/3$ the ratio of test to training providers; p-values are Holm-adjusted over all comparisons. With five rotations these tests have little power, so we also report a provider-level bootstrap of the difference in AP from the best method (1,000 resamples within each rotation's test fold and label, trained models held fixed); it reflects which providers happen to be tested, not retraining variability. To gain power, we repeated the five-fold rotation four more times with reshuffled folds (25 rotations in all) for every method and compared each with GB-P by the same corrected test with $J = 25$ (24 degrees of freedom) [52, 53]. We also re-scored the stored test predictions of those 25 rotations on the group-fraud flags, under a plan committed before any score was computed: *group AP* ranks flagged providers against non-fraud providers with the other fraud providers excluded, the analogue of ring AP, and *individual AP* ranks the other fraud providers against non-fraud providers. Because the tuning split shares providers with the test folds (18–22% of a test fold's providers were in its validation set and 59–62% in its training set), we also re-selected every method's configuration inside each rotation of the first repetition, with the same budget and using only that rotation's validation fold (nested selection; the hybrids inherit their base GNN's choice). Both analyses were pre-specified in the second round of experiments, which covered three of the GNNs in the repeated rotations; the third and fourth rounds added the other GNNs and the hybrids (Section 4.5).

6 Results on the Controlled Benchmark

Each test period contains on average 4,943 claims, of which 384 belong to four unseen rings and 133 are isolated fraud, so the chance levels of ring AP and isolated AP are

0.080 and 0.029. Table 1 reports ring AP for every method and regime over the twelve seeds, Fig. 3 shows the main methods with 95% intervals, and Fig. 4 summarises paired differences. Table 2 averages the paired differences over the nine regimes, with the seed as the unit (eleven degrees of freedom). Recall when the top 5% of claims are investigated cannot exceed 0.479, because fraud makes up 10.5% of test claims, and every method with aggregate features or a graph reached 0.43–0.48 in every regime; at this budget the methods differ little (supplementary results), so we compare them by AP, which covers the whole ranking.

Table 1 Ring-fraud detection on unseen test rings: average precision of ring versus legitimate claims (chance level 0.08), mean over twelve worlds per regime. Bold: best per regime. Standard deviations are in the supplementary results files.

	$\delta = 0$			$\delta = 1$			$\delta = 2$		
Method / s	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
GB-C	0.082	0.083	0.084	0.327	0.327	0.324	0.807	0.809	0.807
MLP-C	0.089	0.083	0.088	0.345	0.349	0.352	0.800	0.804	0.805
LR-CA	0.903	0.871	0.856	0.964	0.955	0.947	0.989	0.986	0.986
RF-CA	0.877	0.851	0.847	0.947	0.928	0.937	0.984	0.982	0.984
GB-CA	0.919	0.896	0.886	0.953	0.936	0.938	0.986	0.985	0.986
GB-CAG	0.924	0.970	**0.990**	0.949	0.976	0.992	0.989	0.993	0.996
SAGE	0.943	0.957	0.971	0.974	0.981	0.990	0.993	0.992	0.992
RGCN	0.935	0.950	0.955	0.968	0.979	0.984	0.994	0.994	0.995
HAN	0.900	0.899	0.921	0.958	0.961	0.967	0.989	0.987	0.988
Simple-HGN	0.917	0.947	0.953	0.968	0.978	0.981	0.992	0.992	0.993
HGT	0.923	0.933	0.955	0.961	0.976	0.977	0.993	0.993	0.993
BWGNN	0.892	0.902	0.914	0.958	0.961	0.963	0.988	0.988	0.990
CARE-GNN	0.884	0.872	0.867	0.955	0.951	0.952	0.990	0.987	0.988
SAGE+AG	**0.962**	**0.979**	0.988	**0.974**	**0.986**	0.995	0.994	**0.996**	**0.997**
HGT+AG	0.959	0.976	0.990	0.973	0.984	**0.996**	**0.994**	0.995	0.997
Overlap score	0.283	0.906	0.982	0.305	0.903	0.980	0.317	0.896	0.978

6.1 Claim features alone do not reveal rings

When ring claims are drawn like legitimate claims ($\delta = 0$), models that see only the claim's own attributes perform at chance (GB-C 0.082–0.084, MLP-C 0.083–0.089). As δ grows their ring AP rises to about 0.33 ($\delta = 1$) and 0.81 ($\delta = 2$). This confirms that the benchmark separates the two sources of evidence as intended.

6.2 At default ring volume, entity aggregates capture most ring evidence

Adding per-period aggregates of the claim's patient, doctor and hospital (GB-CA) raises ring AP at $\delta = 0$ to 0.886–0.919. This is a consequence of the generator's default ring volume rather than a general property of collusion: ring patients file almost six times as many claims as other patients and see more doctors, and ring doctors file

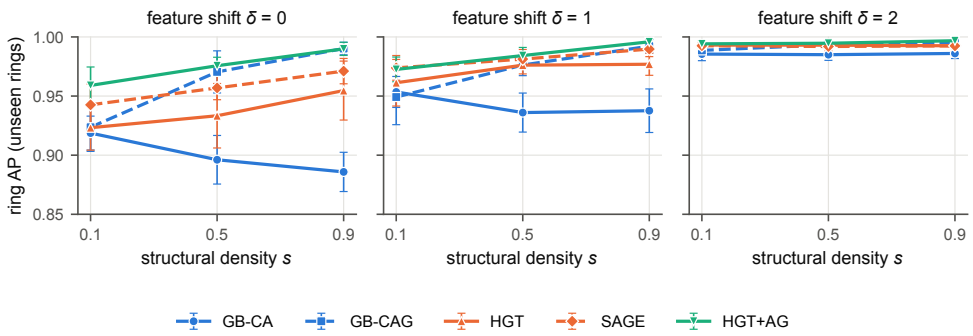


Fig. 3 Ring AP on unseen test rings by feature shift δ and structural density s (mean and 95% t-interval over twelve worlds). GB-CA: gradient boosting on claim and aggregate features; GB-CAG: plus engineered graph features; HGT+AG: HGT whose claim nodes also receive the A and G features. Feature-only models (GB-C, MLP-C) lie far below the plotted range (Table 1).

bursts of claims (Section 3.1). Simple counts capture these signals in every regime; when ring volume is reduced to four claims per month, aggregates detect almost nothing (Section 6.6). The remaining gap at default volume is where coordination structure matters.

6.3 Coordination structure helps only when the ring block is dense

Engineered graph features add little at low density but substantially at moderate and high density: at $\delta = 0$, GB-CAG improves on GB-CA by $+0.005$ ($s = 0.1$, n.s.), $+0.074$ ($s = 0.5$) and $+0.104$ ($s = 0.9$; both Holm $p < 0.001$). The unsupervised overlap score tells the same story: it is weak at $s = 0.1$ (0.283) but reaches 0.906 and 0.982 at $s = 0.5$ and 0.9 without any training. GNNs show the same pattern: SAGE improves on GB-CA by $+0.085$ at $\delta = 0$, $s = 0.9$ (Holm $p < 0.001$). Averaged over the nine regimes, GB-CAG, the hybrids and four of the five general-purpose GNNs outperform GB-CA (GB-CAG $+0.033$, SAGE $+0.034$, RGCN $+0.030$, Simple-HGN $+0.026$, HGT $+0.024$, hybrids $+0.043$ and $+0.042$; Holm $p < 0.001$ with the seed as unit), whereas HAN ($+0.010$), BWGNN ($+0.008$) and CARE-GNN (-0.004) do not significantly (Holm $p \geq 0.059$). When ring claims are strongly anomalous ($\delta = 2$), all methods that see claim and aggregate features exceed 0.98 and the differences become negligible.

6.4 GNNs do not outperform gradient boosting with graph features

The central comparison is between message passing and the same structural information supplied as engineered features. Averaged over the regimes, heterogeneous GraphSAGE matches GB-CAG ($\Delta = +0.001$, 95% CI $[-0.003, 0.006]$), RGCN, Simple-HGN and HGT have small negative point estimates (-0.003, -0.006 and -0.008; Holm $p \geq 0.18$), and HAN is significantly worse (-0.023, Holm $p = 0.002$;

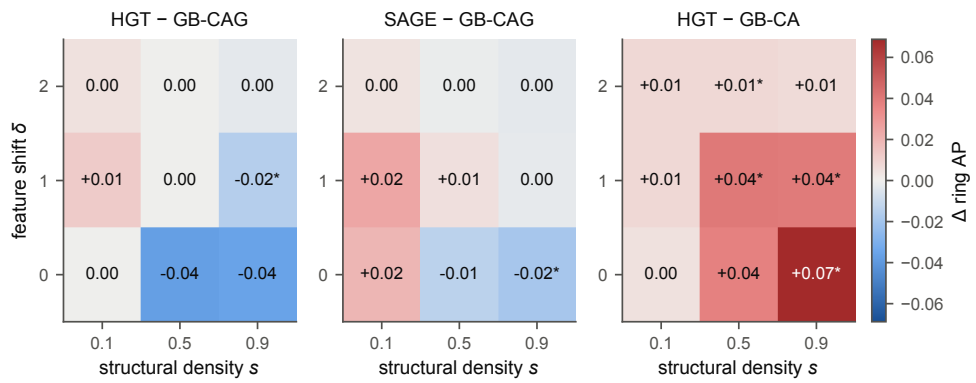


Fig. 4 Mean paired difference in ring AP (twelve worlds per cell). Left and middle: HGT and SAGE versus gradient boosting with engineered graph features. Right: HGT versus gradient boosting without graph features. *Holm-adjusted $p < 0.05$ within the nine-regime family.

Table 2). The two GNNs designed for fraud detection were worse still: BWGNN by -0.025 and CARE-GNN by -0.037 (Holm $p < 0.001$). Every upper CI limit is at most $+0.006$ ($+0.009$ with Bonferroni-adjusted intervals over the seven pure GNNs), so under this tuning the data exclude a pure-GNN advantage much larger than that. Within single regimes, a pure GNN was significantly better than GB-CAG once: RGCN at $\delta = 2$, $s = 0.1$ ($+0.005$, Holm $p = 0.007$), where every method exceeded 0.98. The general-purpose GNNs were significantly worse in eight method–regime pairs, mostly for dense rings and for HAN, and BWGNN and CARE-GNN each in seven of the nine regimes, all but the sparse rings at $\delta = 1$ and 2 (Holm-adjusted within each nine-regime family). Both fraud-specialised models run, like HAN, on claim-only relation graphs, and HAN was the weakest general-purpose GNN, so part of their shortfall may come from that representation rather than from their fraud-specific mechanisms; we did not test this.

The hybrids, which add message passing on top of the engineered features, gained a small but statistically significant amount over GB-CAG: $+0.010$ $[0.005, 0.015]$ for SAGE+AG (Holm $p = 0.006$) and $+0.009$ $[0.004, 0.015]$ for HGT+AG ($p = 0.016$). The gain was largest for sparse rings whose claims look legitimate ($\delta = 0$, $s = 0.1$: $+0.038$ and $+0.035$, Holm $p = 0.007$ and 0.018), where the engineered features add least, and a pre-declared test on fresh seeds confirmed it at low ring volume (Section 6.6). The gain is, however, of the order of the variation between training runs. Averaged over three training seeds (seeds 1–6, $\delta = 0$), SAGE+AG's advantage fell from $+0.013$ to $+0.009$ over the three densities and from $+0.043$ to $+0.030$ for sparse rings, neither significant with six seeds; because the tuning used the same training seed as the reported runs, part of the gain may be specific to it (Online Resource 1, Section S6). Re-selecting among the same candidates in the sparse and dense $\delta = 0$ regimes changed no method's ring AP significantly, and no GNN or hybrid then exceeded GB-CAG significantly (Section S7). Message passing therefore carries little information beyond our engineered features, most of it for sparse rings, and it does not replace them.

Table 2 Paired difference in average precision versus GB-CAG, averaged over the nine regimes: mean $\Delta \pm$ half-width of the 95% t-interval over the twelve generator seeds, and Holm-adjusted paired t-test p (within each metric). The nine regimes of a seed share its legitimate claims, isolated fraud and ring members, so the seed, not the world, is the unit of replication.

	Ring AP		Isolated AP		Overall AP	
Method	$\Delta\pm$ CI	p	$\Delta\pm$ CI	p	$\Delta\pm$ CI	p
SAGE	$+0.001 \pm 0.005$	0.527	-0.097 ± 0.023	<0.001	-0.008 ± 0.004	0.004
RGCN	-0.003 ± 0.004	0.334	-0.109 ± 0.028	<0.001	-0.012 ± 0.004	<0.001
HAN	-0.023 ± 0.010	0.002	-0.138 ± 0.015	<0.001	-0.030 ± 0.007	<0.001
Simple-HGN	-0.006 ± 0.007	0.188	-0.093 ± 0.021	<0.001	-0.014 ± 0.004	<0.001
HGT	-0.008 ± 0.008	0.188	-0.124 ± 0.018	<0.001	-0.017 ± 0.006	<0.001
BWGNN	-0.025 ± 0.008	<0.001	-0.113 ± 0.034	<0.001	-0.035 ± 0.012	<0.001
CARE-GNN	-0.037 ± 0.011	<0.001	-0.229 ± 0.029	<0.001	-0.059 ± 0.012	<0.001
SAGE+AG	$+0.010 \pm 0.005$	0.006	-0.087 ± 0.022	<0.001	0.000 ± 0.004	0.985
HGT+AG	$+0.009 \pm 0.005$	0.016	-0.113 ± 0.022	<0.001	-0.003 ± 0.004	0.374
MLP-C	-0.563 ± 0.018	<0.001	-0.025 ± 0.016	0.006	-0.407 ± 0.022	<0.001

6.5 GNNs are worse on isolated fraud

Isolated fraud shows the opposite pattern (Table 3). Every GNN is significantly worse than GB-CAG, by 0.093 (Simple-HGN) to 0.229 (CARE-GNN) AP averaged over regimes (Holm $p < 0.001$ with the seed as unit), with CARE-GNN, HAN and HGT the weakest. The MLP control, which sees the claim's own attributes and the static attributes of its patient, doctor and hospital—the GNNs' inputs without degree encodings or message passing—comes within 0.025 of GB-CAG ($p = 0.006$). A likely cause is that aggregating a claim's representation with those of its neighbours, which are overwhelmingly legitimate, blurs the claim's own anomaly. The design does not isolate this cause: the MLP also lacks degree encodings, and early stopping on overall AP, where ring claims are most of the positives, may favour rings. The hybrids, whose claim nodes also see the engineered ratios, recover little of the gap (-0.087 and -0.113). On all fraud combined (overall AP), GB-CAG has higher AP than every pure GNN, by 0.008–0.059 (all Holm $p \leq 0.004$), and does not differ from the hybrids ($p \geq 0.37$).

6.6 When ring members are not unusually active

At default volume, ring members stand out by how many claims they file in every regime (Section 3.1). To test which conclusions depend on this, we repeated the three $\delta = 0$ regimes with ring volume reduced from 16 to 4 claims per month and everything else unchanged. In a test period a ring patient then files 3.7 claims, about the 95th percentile of other patients, a ring doctor 20 claims against 16, and rings make up 2.1% of the non-isolated claims, which is also the chance level of ring AP (Table 4). All 16 methods were run with the hyperparameters of the main experiment and again after re-selecting among the same candidates on low-volume development worlds; Section 6.8 searches more widely.

Table 3 Isolated-fraud detection: average precision of isolated-fraud versus legitimate claims (chance level 0.03), mean over twelve worlds per regime. Bold: best per regime.

	$\delta = 0$			$\delta = 1$			$\delta = 2$		
Method / s	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
GB-C	0.483	0.494	0.482	0.574	0.566	0.556	0.651	0.644	0.640
MLP-C	0.529	0.529	0.542	**0.611**	0.607	0.608	0.648	0.656	0.652
LR-CA	0.342	0.299	0.282	0.553	0.538	0.516	**0.668**	0.657	**0.655**
RF-CA	0.553	0.522	0.492	0.592	0.566	0.558	0.629	0.611	0.607
GB-CA	0.508	0.518	0.515	0.577	0.560	0.564	0.651	0.619	0.625
GB-CAG	**0.559**	**0.604**	**0.627**	0.609	**0.609**	**0.626**	0.664	**0.660**	0.651
SAGE	0.414	0.418	0.493	0.524	0.528	0.547	0.611	0.603	0.602
RGCN	0.408	0.440	0.472	0.503	0.511	0.504	0.603	0.597	0.594
HAN	0.390	0.406	0.411	0.476	0.477	0.477	0.584	0.573	0.572
Simple-HGN	0.403	0.412	0.497	0.548	0.516	0.525	0.634	0.625	0.615
HGT	0.339	0.387	0.449	0.509	0.486	0.503	0.618	0.611	0.594
BWGNN	0.431	0.424	0.458	0.503	0.510	0.512	0.590	0.590	0.577
CARE-GNN	0.325	0.249	0.255	0.401	0.418	0.403	0.496	0.494	0.507
SAGE+AG	0.423	0.480	0.569	0.477	0.523	0.577	0.586	0.591	0.605
HGT+AG	0.352	0.447	0.566	0.421	0.499	0.557	0.557	0.587	0.608
Overlap score	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027

With the hyperparameters of the main experiment, three results change. First, per-entity aggregates no longer detect rings: GB-CA reached a ring AP of 0.049–0.055, barely above chance. Second, coordination structure becomes the main source of evidence for moderately and densely linked rings: GB-CAG reached 0.409 and 0.570 at $s = 0.5$ and 0.9, 0.36 and 0.51 above GB-CA (Holm $p \leq 0.001$), against 0.066 at $s = 0.1$, where no method was far above chance. Third, the pure GNNs recovered little of this structure: at $s \geq 0.5$ their ring AP was 0.051–0.151, at most 0.07 above GB-CA averaged over densities, and 0.23–0.29 below GB-CAG (Holm $p \leq 0.002$ with the seed as unit). One result does not change: no method had significantly higher ring AP than GB-CAG at any density, and the hybrids, which receive the engineered features, matched it (SAGE+AG +0.001, 95% CI $[-0.059, 0.061]$; HGT+AG −0.041 $[-0.098, 0.015]$). On isolated fraud the GNNs were again below GB-CAG, by 0.02–0.17, significantly only for CARE-GNN with six seeds (Holm $p = 0.013$).

Re-tuning on low-volume development worlds changed the selected configuration of most methods (HAN, RGCN and Simple-HGN kept theirs and reproduced their results exactly) but not these conclusions (Table 4, right). GB-CAG improved slightly (0.445 and 0.600 at $s = 0.5$ and 0.9, 0.39 and 0.55 above GB-CA), the pure GNNs stayed between 0.050 and 0.162 at $s \geq 0.5$ and 0.26–0.32 below GB-CAG averaged over densities (Holm $p \leq 0.001$), and SAGE+AG matched GB-CAG on average (+0.009 $[-0.048, 0.065]$). The only significant advantage over GB-CAG in either setting was that of SAGE+AG for sparse rings after re-tuning ($s = 0.1$: 0.129 against 0.078, Holm $p = 0.035$), where every method was close to chance. It echoes the default-volume result, where the hybrids' largest gain was also for sparse rings, which suggests that

Table 4 Ring AP on unseen test rings at $\delta = 0$ when ring volume is reduced from 16 to 4 claims per month (mean over six worlds per cell). At low volume a ring patient files about as many claims as the 95th percentile of other patients and a ring doctor 1.3 times as many as other doctors (default: 5.7 and 2.5 times the averages), and the chance level of ring AP falls from 0.080 to 0.021. Frozen: hyperparameters tuned at default volume; re-tuned: the tuning protocol repeated on low-volume development worlds (hybrids inherit their base configuration). Bold: best per column.

	Default volume			Low volume, frozen			Low volume, re-tuned		
Method / s	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
GB-C	0.084	0.086	0.087	0.024	0.023	0.022	0.023	0.023	0.022
MLP-C	0.088	0.087	0.095	0.024	0.024	0.024	0.024	0.023	0.024
LR-CA	0.923	0.888	0.868	**0.106**	0.087	0.088	0.090	0.083	0.089
RF-CA	0.875	0.838	0.844	0.057	0.064	0.064	0.062	0.068	0.073
GB-CA	0.925	0.905	0.889	0.049	0.052	0.055	0.047	0.052	0.051
GB-CAG	0.930	0.983	**0.990**	0.066	**0.409**	**0.570**	0.078	**0.445**	**0.600**
SAGE	0.938	0.963	0.976	0.074	0.108	0.131	0.065	0.112	0.162
RGCN	0.933	0.953	0.954	0.066	0.112	0.142	0.066	0.112	0.142
HAN	0.908	0.906	0.921	0.062	0.065	0.058	0.062	0.065	0.058
Simple-HGN	0.915	0.960	0.957	0.074	0.133	0.151	0.074	0.133	0.151
HGT	0.910	0.948	0.949	0.046	0.079	0.098	0.050	0.082	0.108
BWGNN	0.904	0.918	0.924	0.065	0.076	0.079	0.068	0.073	0.070
CARE-GNN	0.882	0.863	0.868	0.057	0.051	0.053	0.047	0.050	0.058
SAGE+AG	**0.973**	**0.985**	0.983	0.103	0.379	0.566	**0.129**	0.420	0.600
HGT+AG	0.964	0.984	0.990	0.074	0.324	0.523	0.074	0.301	0.477
Overlap score	0.254	0.937	0.985	0.057	0.194	0.281	0.057	0.194	0.281

message passing on top of the engineered features helps most where those features are weakest.

Because this advantage was found by exploring seeds 1–6, we tested it once more on twelve fresh seeds (7–18), with the comparison, test and decision rule fixed beforehand (Online Resource 1, Section S4). SAGE+AG again exceeded GB-CAG, by +0.024 $[0.005, 0.043]$ ($p = 0.016$; nine of twelve seeds), about half the exploratory estimate. That test used one training seed, the one used for tuning, so a further pre-declared test repeated it with two more. On each world's mean over the three training seeds the difference was +0.032 $[0.012, 0.052]$ ($p = 0.005$; eleven of twelve seeds), which meets the rule fixed beforehand for robustness to the training seed. The advantage is real but small, and both methods stayed close to chance (0.136 and 0.104 against 0.021).

Figure 5 traces the transition over four ring volumes (16, 8, 4 and 2 claims per month; seeds 1–6, hyperparameters of the main experiment; full results in Online Resource 1, Table S3). The value of coordination structure is not monotonic in volume. At 16 claims per month aggregates already find most rings and graph features add 0.06 AP averaged over densities. At 8, where ring patients file about as many claims as the 99th percentile of other patients, aggregates fall to 0.31–0.42 and graph features add 0.42 (0.51 and 0.61 for moderately and densely linked rings); at 4 they add 0.30; and at 2 claims per month, one claim per ring member, no method is far above chance. Pure GNNs fall further behind GB-CAG as volume decreases, by 0.01–0.10 AP at 16,

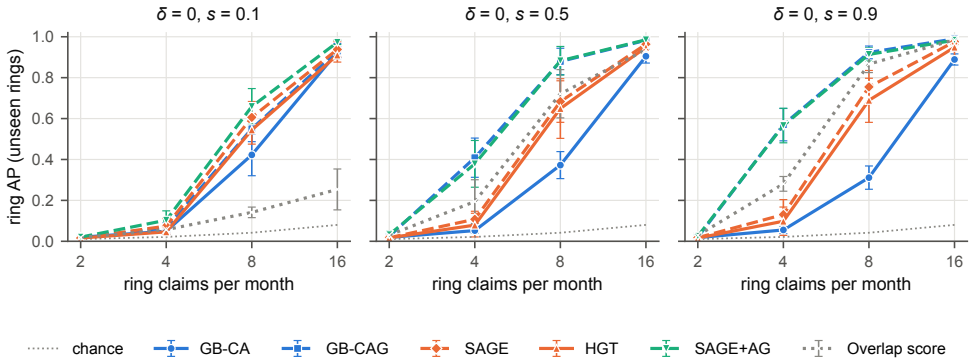


Fig. 5 Ring AP on unseen test rings at $\delta = 0$ against ring volume (claims per ring per month; 16 is the default), mean and 95% t-interval over six worlds (seeds 1–6), hyperparameters of the main experiment. Dotted line: chance level. Structure adds most at intermediate volume; at two claims per month every method is close to chance. GB-CAG is largely hidden behind SAGE+AG, with which it nearly coincides.

0.10–0.43 at 8 and 0.23–0.29 at 4 claims per month, whereas the hybrids stay within 0.05 of it at every volume.

When ring claims were also shifted at volume 4 ($\delta = 1$ and 2; Online Resource 1, Section S5), the moderately and densely linked rings at $\delta = 1$ behaved as at $\delta = 0$: graph features added 0.29–0.30 AP (Holm $p < 0.01$) and the pure GNNs were 0.11–0.29 below GB-CAG. For sparse rings, where graph features added little, SAGE and HAN at both shifts and RGCN at $\delta = 2$ outperformed GB-CAG by 0.07–0.18 (Holm $p \leq 0.030$ within each method's three densities), the only cells of the main and reduced-volume experiments in which a pure GNN did so by more than 0.01 AP. Logistic regression on claim and aggregate features, which sees no graph, outperformed GB-CAG there as well (+0.11 and +0.06), which suggests that the advantage lies in learning a weak claim-level shift from few ring claims rather than in detecting coordination. Pooled over densities, no method outperformed GB-CAG at either shift.

6.7 Other ring shapes and sizes

The engineered features were designed for the default ring, a block of four doctors and twelve patients. We therefore also ran every method on four other shapes and sizes at $\delta = 0$, each at the default per-member volume and at half of it (seeds 1–6; Online Resource 1, Section S8, Table S7): a hub-and-spoke ring; a distributed ring of eight doctors and 24 patients, each patient linked to two ring doctors; and a small (2 doctors, 6 patients) and a large (8 and 24) block. No pure GNN had significantly higher ring AP than GB-CAG for any shape or volume, and pure GNNs were significantly below it in 32 of 56 comparisons, mostly for the hub and the large block (Holm-adjusted over the methods of each shape and volume). Graph features added to aggregates for every shape at half volume (+0.13 to +0.49) and for the hub and the large block also at full volume, so they captured structure they were not designed for. The only significant advantage over GB-CAG was that of SAGE+AG for the distributed ring

(+0.014 and +0.093 at full and half volume, Holm $p \leq 0.023$), whose links are sparser than those of any block in the main grid: as for sparse blocks, message passing on top of the engineered features helped where those features were least informative.

6.8 A wider hyperparameter search

Every earlier tuning re-selected among the same ten candidates per method (Section 4.3). A pre-declared fifth round therefore searched more widely for the four strongest general-purpose GNNs, and gave GB-CAG the same budget (Online Resource 1, Section S15 and Table S12). Each method received 30 candidates, 29 of them new, from a wider grid (up to four layers and 128 hidden units, learning rates down to 3×10^{-4}). Selection and early stopping used validation ring AP, the primary metric, over up to 300 epochs, on development worlds drawn at the evaluation regime. The three cells were the two denser regimes at volume 4, where the GNNs' deficit was largest, and the densest regime at default volume.

The deficit survived. At volume 4 every GNN stayed far below GB-CAG, by 0.26–0.32 at $s = 0.5$ and 0.39–0.56 at $s = 0.9$, with every upper 95% limit below -0.10 (Holm $p \leq 0.014$), so by the pre-declared rule the conclusion stands in both cells. The search raised SAGE at $s = 0.9$ by 0.063 (Holm $p = 0.029$) and changed no other method significantly; the best GNN result, 0.206 for Simple-HGN at $s = 0.9$, compares with 0.593 for GB-CAG. At default volume every GNN remained below GB-CAG, by 0.008–0.063, significantly for RGCN and Simple-HGN (Holm $p = 0.028$). One selection failed. HGT's chosen configuration at volume 4 and $s = 0.9$ reached its best validation score within the first five epochs in five of six worlds, so early stopping ended training before it learned, and its ring AP was 0.029, close to chance. Its main configuration reached 0.098, no closer to GB-CAG. Thirty candidates on two development worlds is still a modest search, and other architectures or training schemes might do better, but the gap at low volume is several times larger than anything the search changed.

6.9 Routing claims to a GNN adds little beyond combining scores

In the design usually proposed, GB-CAG filters and the GNN decides (re-rank). The routed system matched GB-CAG (GB-CAG→HGT ΔAP $= -0.001$, 95% CI $[-0.006, 0.003]$; GB-CAG→SAGE +0.001), and escalating more claims to HGT lowered both ring and isolated AP, because HGT ranks both less well than GB-CAG. Ordering escalated claims by the mean of the two models' percentiles did better: GB-CAG→SAGE gained +0.006 [0.003, 0.010] (Holm $p = 0.042$, positive for all six seeds), but validation then chose to fuse *every* claim in 30 of the 54 worlds, so the gain comes from combining two rankings, not from routing a subset of claims. At the ring level all methods reached the ceiling: every test ring had at least one claim among the top 5% of claims (Online Resource 1, Section S2, Table S2 and Fig. S1).

6.10 Cost

Median training time per world was 3 s for GB-CAG and 39–144 s for the general-purpose GNNs, 13–50 times longer (laptop CPU with several jobs in parallel, so the

times are indicative; the fraud-specialised GNNs ran later under a different load and are not compared), and scoring a whole audit period took at most 0.25 s for any method. On an idle single thread, scoring one claim took a median of 2.1 ms with GB-CAG on a precomputed feature row and 10.7 ms with HGT on the claim's exact three-hop computation subgraph. These figures exclude keeping aggregates, graph features and degree counts up to date as claims arrive (details in Online Resource 1, Section S9).

7 Mechanism and Memorisation

7.1 What isolated and ring fraud look like

Fig. 6 shows the test period of one world (seed 1, $\delta = 0$) with the outcomes of GB-CAG and HGT at their validation-selected thresholds. An isolated fraud claim sits on an ordinary patient–doctor edge, so nothing in the graph distinguishes it and only its own attributes can. A ring at $s = 0.1$ links each recruited patient to one or two of the four ring doctors, a pattern that resembles ordinary shared care, whereas at $s = 0.9$ the same members form an almost complete block. The two worlds share the recruited ring members and every legitimate and isolated-fraud claim, and differ only in which ring links exist. In this world GB-CAG flagged 81 and 91 of the ring's 96 claims at $s = 0.1$ and $s = 0.9$, and HGT 62 and 94: the graph model gains most where the block is dense, as in the averages of Table 1. The figure illustrates one world only; the quantitative statements rest on all 108 worlds of the main grid.

7.2 What the models need

Table 5 reports ablations in the regime where structure matters most ($\delta = 0$, $s = 0.9$), at the default ring volume (with 95% intervals in Online Resource 1, Fig. S2). Degree-preserving rewiring—permuting which doctor bills each claim within a period, which keeps every patient's and doctor's claim volume but destroys the ring's co-occurrence block—reduces every graph-aware method to the same level (0.757–0.770), close to gradient boosting without graph features after rewiring (0.748). The GNNs therefore do exploit coordination structure, and once it is removed nothing distinguishes them from a model that sees volume alone.

Depth matters: with a single message-passing layer, ring AP falls by 0.135 (SAGE) and 0.161 (HGT); two and three layers perform alike. Removing the direct patient–doctor relation costs little (-0.016 and -0.047, not significant after correction), because two-hop paths through claim nodes carry the same information. The largest effect is specific to HGT: without degree encodings its ring AP drops from 0.949 to 0.569 (-0.379, Holm $p = 0.003$), whereas sum-aggregating SAGE changes little (-0.029, n.s.). Softmax attention normalises over neighbours and so cannot count them; supplying counts as input features, or aggregating by summation, restores the signal [42].

Single-factor ablations of HGT (Online Resource 1, Section S10 and Table S8) point the same way. Depth beyond two layers and the number of attention heads made no significant difference, so there is no sign of over-smoothing up to four layers. With

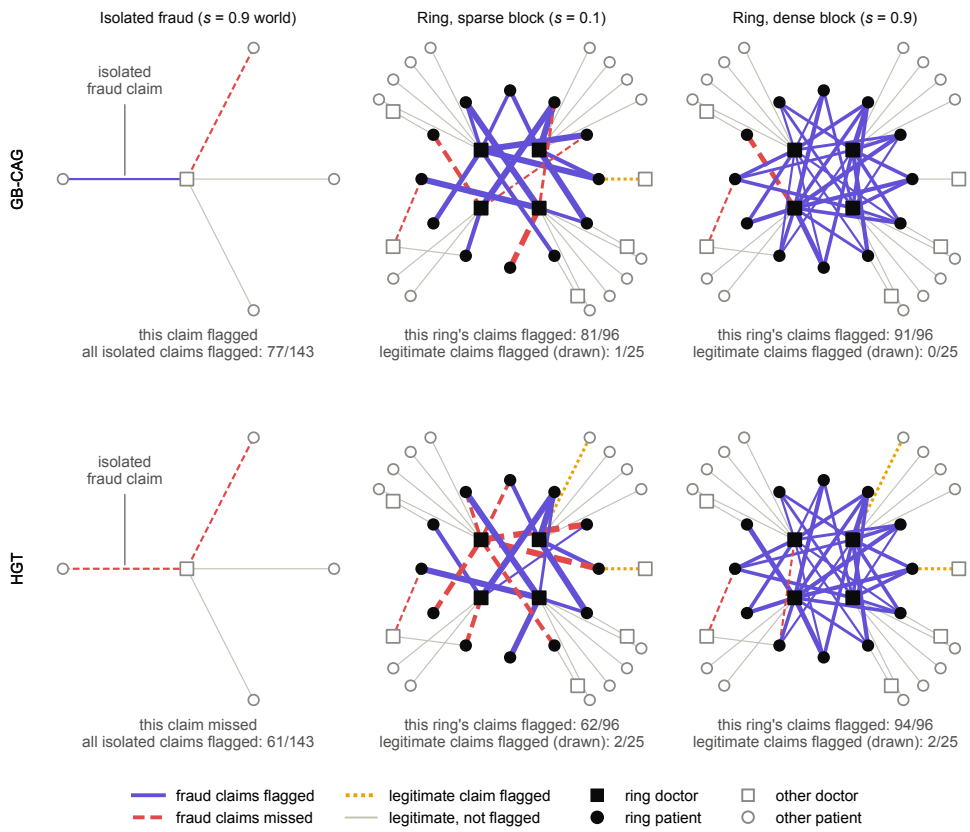


Fig. 6 Isolated and ring fraud in one benchmark world (seed 1, $\delta = 0$; test period), with model outcomes at each model's validation-selected threshold (top: GB-CAG; bottom: HGT). Nodes carry the ground truth (filled: ring member; squares: doctors; circles: patients); edges are patient–doctor pairs with claims in the period, coloured by the outcome for the majority of their claims (line width: number of claims). Left: an isolated fraud claim drawn at random, its patient's doctors and up to ten other patients of its doctor. Middle and right: one test ring, every doctor its patients visit and up to four other patients per ring doctor, at low and high structural density; the two worlds share all recruited members and all legitimate claims. The figure illustrates the generator's typologies; it is not evidence that real rings look alike.

claim attributes set to zero, ring AP was unchanged and isolated AP fell to chance; an HGT that saw only the degree encodings and the graph reached a ring AP of 0.982 ± 0.018, as high as the full model, and detected almost no isolated fraud. HGT's ring detection is therefore structural—counts of who bills whom, propagated along patient–claim–doctor paths—which is the information that the engineered shared-patient features summarise directly. At default volume these counts also carry the ring members' excess claims; Section 6.6 shows that when they do not, the GNNs fall far behind gradient boosting with the engineered features. Permuting groups of inputs of the trained models (Online Resource 1, Section S11) is consistent with this: GB-CAG relied on aggregates when rings were sparse and on the graph features, above all the

Table 5 Mechanism ablations at $\delta = 0$, $s = 0.9$ (six worlds): ring AP (mean ± s.d.) and paired change from the full model, with Holm-adjusted p over every ablation comparison in this table and Table S8 of Online Resource 1, which lists HGT's single-factor variants. The tuned GNNs use three layers.

Method	Variant	Ring AP	Δ vs. full
GB-CA	Full model	0.889 ± 0.026	–
	Degree-preserving rewiring	0.748 ± 0.027	−0.141 ($p = 0.018$)
	+ ring identifier (legacy leak)	1.000 ± 0.000	+0.111 ($p = 0.003$)
GB-CAG	Full model	0.990 ± 0.008	–
	Degree-preserving rewiring	0.757 ± 0.046	−0.232 ($p = 0.002$)
SAGE	Full model	0.976 ± 0.013	–
	Degree-preserving rewiring	0.759 ± 0.034	−0.217 ($p = 0.002$)
	No patient–doctor relation	0.960 ± 0.019	−0.016 ($p = 0.360$)
	No degree encodings	0.947 ± 0.037	−0.029 ($p = 1.000$)
	1 message-passing layer	0.841 ± 0.034	−0.135 ($p = 0.002$)
	2 layers	0.957 ± 0.023	−0.019 ($p = 0.413$)
HGT	Full model	0.949 ± 0.052	–
	Degree-preserving rewiring	0.770 ± 0.029	−0.179 ($p = 0.036$)
	No reverse relations (legacy graph)	0.772 ± 0.024	−0.177 ($p = 0.022$)

normalised triadic closure of the claim, when they were dense, whereas SAGE and HGT relied almost entirely on degree encodings.

A pre-declared experiment then asked what the pure GNNs lack when ring volume is low (Online Resource 1, Section S16 and Table S13). Apart from a degree term, the claim closure counts four-cycles through the claim's patient–doctor edge, a pattern that message passing cannot count in general [37]. At volumes 4 and 8 ($\delta = 0$, $s = 0.5$ and 0.9), giving a GNN only the two closure columns recovered 82–90% of SAGE's gap to GB-CAG and 52–84% of HGT's (gains 0.15–0.36, Holm $p \leq 0.027$). SAGE's remaining gap of 0.02–0.08 was not significant in any cell, with six seeds (Holm $p \geq 0.18$), whereas HGT stayed significantly below GB-CAG at volume 4 (Holm $p \leq 0.033$). The other half of the prediction failed: GB-CAG without those two columns lost only 0.01–0.07 (Holm $p \geq 0.21$) and kept 79–98% of its structure gain. Our pre-declared rule therefore applies, and we do not offer four-cycle counting through the claim as the explanation of the GNNs' deficit. The contrast with the permutation analysis is informative: a trained GB-CAG uses the closure heavily, but a retrained one can do without it. After the fact, we note that the prediction could not have isolated the closure, because four of the five remaining columns are also shared-patient statistics, two of which test directly whether doctors share a second patient; this reading is post hoc and untested. What the experiment does show is narrower: with the same training claims and hyperparameters, two columns of shared-patient information closed most of a gap that the GNNs did not close on their own.

7.3 Two evaluation pitfalls, quantified

The mechanism ablations (Table 5) also quantify two defects that are easy to introduce. Adding the generator's ring identifier to the feature set—a column that

is non-zero only for ring claims—raises gradient boosting from 0.889 to a perfect 1.000. Building the heterogeneous graph with one-directional relations only (patient→claim→doctor→hospital) leaves claim nodes reachable only from patients, so doctor context never reaches them; HGT then loses 0.177 AP. Both defects produce plausible-looking numbers and neither is caught by aggregate metrics alone; the tests listed in Section 10 detect both.

7.4 Recognising known members is not detecting rings

Table 6 (and Online Resource 1, Fig. S3) compares new rings (emergent protocol) with rings whose members recur from training (persistent protocol). The MLP control, which sees only the claim and the static attributes of its patient, doctor and hospital, rises from chance to 0.692 ($s = 0.1$) and 0.591 ($s = 0.9$) when rings recur: it identifies the members from their attributes. HGT gains 0.056 at $s = 0.1$ (95% CI $[0.029, 0.084]$), and gradient boosting with aggregates gains 0.035–0.063, while the unsupervised overlap score, which cannot memorise, is unchanged. Evaluations in which test rings share members with training rings therefore overstate how well a model detects coordination, by an amount that differs between methods and can reorder them.

Table 6 Ring AP when test rings are new (emergent) versus recurring members of training rings (persistent), $\delta = 0$, six worlds each; gain = persistent − emergent (paired over worlds, 95% t-interval). The overlap score is unsupervised and cannot memorise.

	$s = 0.1$			$s = 0.9$		
Method	new	recurring	gain	new	recurring	gain
MLP-C	0.088	0.692	+0.604 [+0.523, +0.685]	0.095	0.591	+0.496 [+0.416, +0.577]
GB-CA	0.925	0.959	+0.035 [+0.014, +0.055]	0.889	0.952	+0.063 [+0.034, +0.092]
GB-CAG	0.930	0.955	+0.025 [−0.012, +0.062]	0.990	0.994	+0.005 [−0.002, +0.012]
SAGE	0.938	0.969	+0.031 [−0.019, +0.081]	0.976	0.978	+0.002 [−0.015, +0.020]
HGT	0.910	0.967	+0.056 [+0.029, +0.084]	0.949	0.981	+0.032 [−0.017, +0.082]
Overlap score	0.254	0.256	+0.002 [−0.137, +0.142]	0.985	0.991	+0.006 [−0.009, +0.020]

8 Results on the Public Provider-Labelled Data

Table 7 and Fig. 7 report provider-level results over the five rotations (1,082 test providers each, about 101 fraudulent). Gradient boosting on provider aggregates (GB-P) has the highest mean AP (0.704 ± 0.045). Adding label-free graph features (GB-PG, -0.009, corrected 95% CI $[-0.042, 0.023]$) or graph and label-neighbourhood features (GB-PGL, -0.007 $[-0.030, 0.016]$) does not improve it. Every GNN has lower AP than GB-P, by 0.029 (HGT) to 0.083 (RGCN), and the hybrids, which give the GNN the provider and graph features, are no better than the plain GNNs. With five overlapping rotations, no difference from GB-P or GB-PGL is significant after the correction and Holm adjustment over 28 tests, one pair counted twice (smallest Holm $p = 0.15$, SAGE+G versus GB-P); a plain paired t-test, which ignores the overlap, would have called SAGE+G significantly worse (Holm $p = 0.025$). Holding the trained models fixed, the provider-level bootstrap places the AP of every GNN and hybrid, and of GB-PG, below that of GB-P (95% intervals from $[-0.108, -0.055]$ for RGCN to $[-0.048, -0.006]$ for HGT), so for these trained models the shortfall is not explained by which providers happened to be tested. This is descriptive: GB-P was identified as the best method on the same folds, and the fourteen intervals are not adjusted for multiplicity. ROC-AUC lies between 0.927 and 0.942 for all methods and recall among the top 10% of providers between 0.59 and 0.65, so the methods rank providers similarly; the AP differences come from the top of the ranking. Under the same parallel load as before, median training time per rotation was 2–15 minutes for the general-purpose GNNs and hybrids (range 1.4–24 minutes) against 7 s for GB-P; the fraud-specialised GNNs, run later under a different load, are not included.

The two pre-specified checks of this protocol left the picture unchanged (Online Resource 1, Section S13). Over 25 rotations (five repetitions of the five-fold rotation), GB-P again had the highest mean AP (0.693), graph features changed it by -0.007 $[-0.019, 0.006]$ (GB-PG) and -0.006 $[-0.019, 0.007]$ (GB-PGL), the seven GNNs were 0.025–0.092 and the hybrids 0.039–0.063 below it. Only RGCN's deficit was significant after the correction and Holm adjustment (Holm $p = 0.009$; next smallest 0.062, SAGE+G), whereas a plain paired t-test over the 25 rotations would have called all fourteen differences significant. Re-selecting every method's configuration inside each rotation changed AP by at most 0.010 for the tabular models and the MLP and by -0.008 to $+0.036$ for the GNNs and hybrids, none significantly (SAGE chose its shared configuration in every rotation). Contrary to our expectation, the shared selection therefore did not systematically favour the GNNs. With every method selected inside each rotation, GB-P (0.706) remained the best, and every GNN and hybrid was 0.023–0.053 below it (smallest Holm $p = 0.059$).

The graph statistics explain why. Almost every provider (99.5%) shares beneficiaries with other providers (median 30 co-providers), but labels are not assortative through this relation: on average 23.9% of a fraudulent provider's co-providers are labelled fraudulent, against 25.3% for a legitimate provider (both well above the 9.4% base rate because shared beneficiaries concentrate on large providers). Shared physicians show mild assortativity (19.1% versus 14.2%) but connect only 73% of providers, with a median of two co-providers. Fraudulent providers are, however, much larger (median 156 versus 27 claims), a volume signal that provider aggregates capture

Table 7 Provider-level fraud detection on the public provider-labelled data (5-fold rotation; mean ± s.d. over folds). F1 at the validation-selected threshold; Recall@10%: share of all fraudulent providers that rank among the top 10% of providers by score.

Method	AP	ROC-AUC	F1	Recall@10%
LR-P	0.678 ± 0.041	0.940 ± 0.017	0.617 ± 0.030	0.644 ± 0.023
RF-P	0.681 ± 0.049	0.942 ± 0.017	0.624 ± 0.031	0.644 ± 0.025
GB-P	0.704 ± 0.045	0.939 ± 0.020	0.624 ± 0.028	0.636 ± 0.027
GB-PG	0.695 ± 0.040	0.937 ± 0.019	0.600 ± 0.043	0.642 ± 0.024
GB-PGL	0.697 ± 0.042	0.937 ± 0.019	0.595 ± 0.045	0.634 ± 0.019
MLP-P	0.685 ± 0.052	0.940 ± 0.019	0.613 ± 0.028	0.642 ± 0.032
SAGE	0.664 ± 0.047	0.938 ± 0.018	0.605 ± 0.036	0.624 ± 0.046
RGCN	0.621 ± 0.054	0.927 ± 0.019	0.578 ± 0.029	0.593 ± 0.029
HAN	0.666 ± 0.046	0.938 ± 0.018	0.583 ± 0.028	0.636 ± 0.045
Simple-HGN	0.670 ± 0.057	0.939 ± 0.020	0.600 ± 0.037	0.630 ± 0.036
HGT	0.676 ± 0.058	0.939 ± 0.019	0.613 ± 0.051	0.640 ± 0.051
BWGNN	0.673 ± 0.069	0.938 ± 0.019	0.607 ± 0.036	0.644 ± 0.037
CARE-GNN	0.669 ± 0.053	0.940 ± 0.018	0.621 ± 0.043	0.648 ± 0.038
SAGE+G	0.653 ± 0.048	0.937 ± 0.018	0.612 ± 0.040	0.623 ± 0.031
HGT+G	0.648 ± 0.032	0.933 ± 0.014	0.589 ± 0.022	0.615 ± 0.035

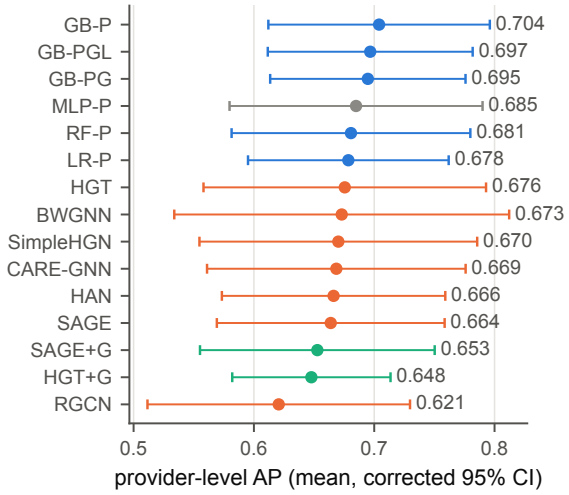


Fig. 7 Provider-level AP on the public provider-labelled data (mean over five rotations; 95% CI from the corrected resampled t-distribution, which accounts for the overlap of the rotations' training sets). Blue: tabular; orange: GNN; aqua: hybrid; grey: MLP.

directly. In the terms of our benchmark, these data resemble the low-density regime, where structure added nothing beyond per-entity aggregates. The labels mark potentially fraudulent providers without identifying rings, so the absence of a graph benefit here does not show that rings are absent from Medicare data; it shows that this public labelling contains no exploitable relational signal for the models tested.

Group-fraud flags.

Re-scoring the same models on the published flags changed little (Online Resource 1, Section S17 and Table S14). GB-P and GB-PG reached a group AP of 0.719 against a chance level of 0.020, and graph features added nothing (GB-PG −0.001, 95% CI $[-0.026, 0.024]$, Holm $p = 1.000$). All nine graph models were below GB-PG, by 0.10–0.21, significantly only for SAGE+G (Holm $p = 0.031$). Both outcomes had been predicted before any score was computed, because the flagged providers were known to be large: a median of 618 claims, against 114.5 for other fraud providers and 27 for non-fraud providers. In the terms of our benchmark these are unusually active members, for whom aggregates suffice. Flagged providers were no more densely linked through shared beneficiaries than other fraud providers (median maximum Jaccard 0.087 and 0.088), but they were assortative through shared physicians: 13.7% of their co-providers were flagged, against 2.9% for other providers. Retraining on the flag itself left gradient boosting level with and without graph features (−0.009, 95% CI $[-0.072, 0.055]$); over five rotations SAGE, RGCN and HGT reached 0.640–0.758 against 0.775 for GB-PG, with intervals too wide to exclude a difference in either direction. The flags record an undocumented expert judgement on claims that may be synthetic, so this is a consistency check. It does not test dense rings of otherwise unremarkable members, where our benchmark found structure most valuable.

9 Discussion

9.1 When does relational structure help?

Table 8 summarises the conditions that emerge from the controlled study. First, the claims must not already be anomalous on their own: when ring claims were strongly upcoded ($\delta = 2$), every model with claim and aggregate features exceeded 0.98 AP and structure became redundant. Second, the value of structure depends on what else distinguishes ring members: at default volume their elevated utilisation, which per-entity counts capture, already identified most rings, whereas at four to eight claims per month aggregates lost most of their power and coordination structure carried most of the evidence. Third, the trace of coordination must be dense enough to distinguish a ring from ordinary shared care: graph information added 0.07–0.10 AP for moderately and densely linked rings at default volume and 0.36–0.61 at reduced volume, but less for sparse rings (+0.005 at default volume, +0.13 at eight and at most +0.03 at four claims per month). Finally, a ring that files about one claim per member leaves no trace that any of our methods could find.

9.2 Why GNNs did not beat engineered graph features

When structure did help, gradient boosting with a handful of shared-patient statistics captured it as well as or better than the best GNN. GNNs designed for fraud detection did not change this: BWGNN, whose band-pass filters are meant to keep anomalies from being smoothed away, and CARE-GNN, which discards dissimilar neighbours, were both below GB-CAG on average and in most regimes. The mechanism analyses

suggest why. At default volume much of the useful signal is *neighbour counting*—how many claims and counterparties each entity has—and attention- or mean-based aggregators cannot count neighbours: HGT lost most of its ring detection without degree encodings, whereas sum-aggregating GraphSAGE barely changed, and an HGT given only degree encodings and the graph detected rings as well as the full model (Section 7). This mirrors the expressiveness argument for sum aggregation [42]. GNNs were also markedly worse on isolated fraud, consistent with neighbourhood averaging diluting claim-level anomalies, and they readily memorise the attributes of training-ring members, which inflates performance when rings recur but does not transfer to new rings. Hybrids that combine message passing with the engineered features gained a small amount over gradient boosting on ring fraud (+0.009 to +0.010 AP, of the order of the variation between training runs), most for sparse rings, where the engineered features are weakest; that is the structure message passing extracts beyond our features, and it comes at 9–40 times the training cost. The one setting in which pure GNNs clearly beat gradient boosting with graph features, sparse rings at reduced volume whose claims were also shifted, is one in which structure added little and a linear model without graph inputs beat it as well.

The ring-volume experiments sharpen this picture and separate two kinds of counting. Neighbour counts are within reach of message passing and, at default volume, carry the ring members' excess claims. Whether a claim's patient and doctor sit in a block of shared patients is a different matter: apart from a degree term the claim closure counts four-cycles, which message passing cannot count in general [37]. Once ring members no longer stood out by volume, the pure GNNs recovered only a small part of the co-occurrence structure that the engineered closure features summarise, with the main hyperparameters, after re-tuning and after a wider search alike, whereas hybrids given those features came close to or matched gradient boosting and, for sparse rings, exceeded it. Message passing over two training periods with about 200 ring claims did not learn the shared-patient pattern that a single engineered feature states directly: at $s \geq 0.5$ the unsupervised overlap score alone (0.194 and 0.281) had higher ring AP than every pure GNN in both settings. Two closure columns closed most of that gap (Section 7.2), so the deficit can be remedied cheaply. We claim no more than that. The limit on counting is a worst-case result and does not show that a trained network could not approximate the count on these data, the few training ring claims may matter as well, and gradient boosting proved not to depend on the closure columns themselves.

One caveat favours the engineered features: we designed them knowing the injected typology, a dense doctor–patient block. Four other shapes and sizes did not overturn the result: the features still captured hub-and-spoke rings and larger and smaller blocks, no pure GNN beat them for any shape, and the hybrid's gain was largest for the distributed ring, whose sparse links the features summarise least well (Section 6.7). Typologies further from those we tried, such as rings that change members over time or that mimic ordinary referral patterns, could still favour learned message passing. These findings are consistent with evidence from other domains that tree ensembles with neighbourhood features are strong graph anomaly detectors [10, 11, 35], with the

same pattern on public Medicare provider data [12], and extend it to heterogeneous patient–provider claim graphs with controlled typologies.

9.3 Why a small fixed benchmark cannot settle the question

A small fixed benchmark can appear to test ring detection without being able to. We re-analysed one, a dataset of 2,100 claims with five injected rings produced by an earlier, simpler generator of ours (Online Resource 1, Section S14). It cannot tell whether relational structure helps: its ring claims were anomalous on their own (gradient boosting on three raw claim attributes reached a ring AP of 1.000), its rings recurred across the split, so every test ring claim was billed by a doctor with labelled ring claims in training, and its test split held only 21 ring claims. Controlling the feature-level signal, testing on new rings and replicating worlds removes all three confounds.

9.4 Public provider-labelled data

On the public provider-labelled data, gradient boosting on provider aggregates had the highest AP, and neither engineered graph features, label-neighbourhood features nor any GNN improved on it, in agreement with earlier comparisons on these data [12, 31]. This held over 25 rotations, where graph features changed GB-P's AP by at most +0.006 to +0.007 (upper 95% limits), and with every model's configuration selected inside each rotation. The labels are not assortative through shared beneficiaries, and fraudulent providers differ mainly in volume, so the data sit in the regime where our benchmark also found no benefit from structure. The same held, as predicted, for the providers that experts flagged as group fraud, who also stand out by volume. Ma et al. [13] reported that their GNN beat baselines on those flags, but its scores are not comparable with ours: it scores claims on a temporal split in which most test-period providers also appear in training. Our results are consistent with the controlled findings but are not an independent confirmation of the dense-ring regime, which requires labelled collusive groups whose members are not otherwise conspicuous; no public labels we found show this. Because the source of the claims and of the labels is undocumented, the result should also not be read as an estimate of how these methods perform on current Medicare claims.

9.5 Practical recommendations

- Start with gradient boosting on claim features, per-entity aggregates and a small set of shared-patient graph features. On our benchmark no pure GNN, including two designed for fraud detection, was better on average or for any ring shape we tried, its one significant per-regime advantage (0.005 AP) came where every method exceeded 0.98, and gradient boosting was 9–50 times cheaper to train than the general-purpose GNNs and hybrids. Adding message passing on top of the engineered features gained about 0.01 AP, most for sparse rings and sparsely linked ring shapes; where rings may be sparse, a hybrid is worth testing, although every method detected sparse rings poorly when their members were not unusually active. Keep a linear model among the baselines as well: where such rings' claims were also mildly anomalous, logistic regression and several GNNs outperformed gradient boosting.

- Do not infer the value of relational structure from rings whose members are unusually active: then aggregates already find them. When they are not, explicit co-occurrence features, or hybrids that receive them, were needed.
- Evaluate on rings (or providers) that do not appear in training, and compare against a no-graph control with identical inputs; otherwise memorisation of known members can masquerade as relational learning. Report uncertainty over the independent units—generated worlds that share data, or cross-validation rotations that share training data, are not independent.
- If a GNN is used, include reverse relations, verify that information reaches the classified nodes, and give attention-based architectures structural (degree) encodings or prefer sum aggregation.
- Keep isolated-fraud detection on a claim-level model; every GNN had lower AP on it than GB-CAG in every regime.
- Do not expect a GNN second stage to pay for itself. Letting a GNN re-rank the claims that gradient boosting flags gained nothing here; averaging the two models' rankings gained at most 0.006 AP, and in an audit setting, where a whole period is scored at once, routing saves little computation (Section 6.9).

For a payer's health information system these results favour a modest design. The aggregates and shared-patient statistics used here are counts and overlaps over one audit period of claims, which a claims database can compute in batch, and the resulting model scored a whole period in a fraction of a second on a laptop CPU (Section 6.10). A GNN adds a graph-learning stack and 13–50 times the training cost of gradient boosting in our runs; a pure GNN brought no gain on average in our benchmark or on the public data, and a hybrid about 0.01 AP, most for sparse rings. Because investigators examine only the highest-ranked cases, such systems should be evaluated at the investigation budget and on providers and rings unseen in training.

9.6 Limitations

The controlled findings hold for our generator. Its distributions are simple, labels are noise-free and immediate, rings are static during their active period and, apart from five shape and size variants at $\delta = 0$ with six seeds, of one shape and size, and ring prevalence (8% of the non-isolated test-period claims at default volume) is far above realistic levels so that each world has enough positives for stable estimates; the ring-volume experiments lower it to 1.1–4.2% but rest on six seeds and cover $\delta > 0$ only at four claims per month. The engineered graph features were designed with knowledge of the injected typology, which favours them over learned message passing (Section 9). We tuned each method once, on development worlds from a single regime ($\delta = 0.5$, $s = 0.5$), and once more for volume 4; re-tuning in two further regimes changed no result significantly, but other regimes were not re-tuned. A wider search of 30 candidates for four GNNs and GB-CAG in three cells left the conclusions unchanged, but it chose on two development worlds, and one selected configuration failed to train on most evaluation worlds. The twelve seeds per regime are the unit of replication: pooled tests have eleven degrees of freedom and cannot resolve differences much below 0.005–0.01 AP, and the persistent-ring, ablation, routing and ring-volume

analyses rest on six seeds. Seeds 7–12 were added after the first results had been seen, under a plan fixed before the new runs; the first-round results remain in the repository. Each model was trained once per world, from the same initial weights as in tuning; a replication with two further training seeds covered five methods at $\delta = 0$ and showed that the GNNs' variation between training runs is comparable to their variation between worlds, and the confirmatory test was repeated over three training seeds. Graph models were trained full-batch on graphs of about 11,000 nodes per period; larger graphs would need sampling-based training [44]. The architecture and reliance analyses were run in one regime ($\delta = 0$, $s = 0.9$) with one model seed per world, and permutation importance can understate a feature group whose information is duplicated in another; these analyses describe what the trained models use, not causal effects of the generator. Routing was evaluated with one first stage, two GNNs and a grid of six escalation fractions, and cost excludes the upkeep of aggregates, graph features and degree counts as claims arrive. On the public data, labels are per provider, the source of the claims and of the labels is undocumented, and the observation window is a single year. The group-fraud flags are expert judgements by an undocumented criterion, so any relational signal in them may be partly built in, and RDPGL itself, whose code is not public, was not run. Even 25 overlapping rotations leave the corrected tests little power for differences of a few hundredths of AP. Each model was trained once per rotation. The two fraud-specialised GNNs are our implementations, adapted to the claim-only relation graphs and full-batch training used here; other fraud GNNs, such as PC-GNN [29], were not compared, and neither fraud GNN was included in the persistent-ring, ablation, routing, training-seed or per-regime-tuning analyses. Substructure-aware GNNs, which are designed to count patterns such as four-cycles, were not tested, so our results do not extend to them. Finally, the benchmark does not model adaptive fraudsters, temporal ring dynamics, which temporal graph models target [54], or realistic coding semantics such as those of synthetic health records [55], and its rings are injected: benchmarks of graph outlier detection assess methods separately on injected and on organic outliers [56], and ours cannot stand in for organic collusion.

10 Reproducibility

All results are produced by one released package from raw inputs. Every run writes a record containing the git commit, a hash of its configuration, the hashes of the generated claim and ground-truth tables, the library versions and the per-claim (or per-provider) test predictions; the tables and figures of this paper are generated from these records by a single script. Automated tests, run in continuous integration, check that (i) no ground-truth column reaches any feature matrix and features are invariant to permuting the labels; (ii) a period's features and graph are unchanged when claims of other periods are perturbed; (iii) emergent rings in different splits share no doctor or patient; (iv) at $\delta = 0$ ring claims are distributed like legitimate claims; (v) features, graphs and labels stay aligned by claim identifier; and (vi) for every GNN, perturbing doctor features (or, for the fraud-specialised GNNs, a claim's relation neighbours) changes claim logits, so relational information actually reaches the classified nodes;

(vii) a routed score depends only on the claim's own two scores and validation statistics; (viii) rewiring the test period leaves the features of other periods unchanged; (ix) scoring a claim on its L-hop computation subgraph reproduces its full-graph score; and (x) the regimes of a seed share their legitimate claims and ring members, the design that makes the seed the unit of pooled tests; and (xi) the default ring shape reproduces the recorded worlds exactly, while the other shapes link ring members as specified. A further check confirms that every table in this article equals the file generated from the run records. The first-round runs were made from a working tree that had not yet been committed, so their records do not pin the code version; retraining GB-CAG, SAGE and HGT with a later commit on all 54 worlds reproduced their stored test predictions (Section 4.4). Later records also store a hash of the research sources at process start. These hashes show that 33 run files of the low-volume and second-round analyses came from source states that match no commit, and further runs are flagged as having uncommitted changes although their Python sources match a commit. We therefore re-ran all 33 files from a copy of the first tagged release: all 227 method runs reproduced the stored AP exactly, 221 with identical test scores and six with scores differing by at most 6×10^{-8} (the log is in the repository). The fifth round ran from one committed, unmodified version of the code, and every round after the first followed an analysis plan committed before its runs. A checklist mapping this study to the REFORMS recommendations [57] and a datasheet for the generator [58] are included in the repository.

11 Conclusion

We asked when relational structure helps healthcare fraud detection and answered it with a benchmark that varies the feature-level signal, density and volume of provider rings independently, tests on rings never seen in training, and checks the conclusions on public provider-labelled data. Structure helped most when rings formed moderately or densely linked doctor–patient blocks, and it mattered most when ring members were not otherwise unusual: then per-entity aggregates detected little, while a ring filing about one claim per member could not be found at all. Gradient boosting with a handful of engineered shared-patient features matched or outperformed five heterogeneous and two fraud-specialised GNNs on average, and for every ring shape we tried, at a fraction of the cost, and at reduced ring volume it detected far more of the densely linked rings than any pure GNN, also after a wider hyperparameter search; adding message passing on top of those features gained about 0.01 AP, of the order of the variation between training runs, most for sparse rings, and routing high-risk claims to a GNN gained little. Some pure GNNs outperformed it, but only for sparse rings with mildly anomalous claims at reduced volume, where a linear model without graph inputs did so too. GNNs generalised to new rings only with reverse relations, sufficient depth and counting ability, degraded isolated-fraud detection, and could memorise known ring members in ways that inflate conventional evaluations. A small fixed benchmark whose rings are anomalous in their features, recur across the split and contribute few test claims cannot separate relational learning from claim-level anomaly detection or from recognising known members. On public provider-labelled

data no graph-based method improved on tabular aggregates, as earlier work on these data also found, and the same held for providers that experts flagged as group fraud, who stand out by volume. These conclusions rest on a simulator whose engineered features were designed for its default typology, although they also captured four other ring shapes and sizes. We recommend that graph models for claims fraud be justified against a graph-augmented tabular baseline and a no-graph control, on entities unseen in training, with uncertainty computed over independent units. The benchmark, code and tests are publicly available to support such comparisons; the most valuable next step is real data with labelled collusive schemes whose members are not otherwise conspicuous.

Supplementary information. Online Resource 1 (PDF) contains the selected hyperparameters, the two-stage routing evaluation, the full ring-volume dose-response, the confirmatory test of the hybrid result, three robustness checks (feature shift at reduced ring volume, training-seed replication and per-regime tuning), results for other ring shapes and sizes, the cost measurements, the single-factor ablations of HGT, the input-reliance analysis, figures of the mechanism ablations and of new versus recurring rings, repeated cross-validation and per-rotation tuning on the public data, the re-analysis of the small fixed benchmark, the wide hyperparameter search, the closure-column experiment and the analysis of the group-fraud flags. The repository contains the per-world results (including standard deviations, F1, ROC-AUC and recall at a fixed investigation budget for every regime and method), the tuning logs and the run records from which every table and figure is generated. The per-claim and per-provider test predictions (about 30 MB) are not tracked in the repository; the released code regenerates them.

Statements and Declarations

- **Funding.** The authors did not receive support from any organization for the submitted work.

- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** Not applicable: the study uses synthetic data and a public dataset released under a CC0 licence, in which beneficiaries, providers and physicians are identified only by pseudonymous codes; no human participants were involved and no new data were collected.
- **Consent for publication.** Not applicable: the manuscript contains no individual person's data in any form; it uses synthetic data and a public dataset with pseudonymous identifiers only.
- **Data availability.** The synthetic benchmark, at every ring volume, shape and size studied, is generated by the released code from fixed seeds. The public provider-labelled data are available from Kaggle [51] under CC0, and the group-fraud flags from https://github.com/finint/RDPGL (commit `6e0151d`); file checksums are listed in the repository.
- **Code availability.** Code, configurations, tests and result records are available at https://github.com/akshay-hudev/Fraud-Guard (directory `research/`; the version

described here is tagged `hiss-submission-v2`) under the MIT licence. An archived release with a DOI will be provided upon acceptance.

- **Author contribution.** H M Akshay: conceptualization, methodology, software, investigation, formal analysis, data curation, visualization and writing—original draft. Gaurav S B, Eshita Narayana and Harshita Srivastava: methodology, investigation, formal analysis and writing—original draft. Malini M Patil: supervision and writing—review and editing. All authors read and approved the final manuscript.

Appendix A Generator parameters

Table 8 When relational structure helps: a summary of the controlled results ($\delta = 0$ unless stated). Structure gain: GB-CAG minus GB-CA in ring AP. Volume: ring claims per month (default 16).

Setting	Main evidence for rings	Structure gain	What works
Ring claims anomalous ($\delta = 2$)	Claim attributes	None at volume 16 (all methods with aggregates > 0.98) or 4 (+0.04, n.s.)	Any tabular model with aggregates
Members unusually active (volume 16), sparse rings ($s = 0.1$)	Member volume, repeated doctor–patient pairs	+0.005	GB-CA(G); hybrids add +0.035 to +0.038
Members unusually active (volume 16), dense rings ($s \geq 0.5$)	Volume and shared patients	+0.07 to +0.10	GB-CAG or hybrids; pure GNNs no better
Members moderately active (volume 8), sparse rings	Volume and shared patients	+0.13	GB-CAG, hybrids and general-purpose GNNs other than HAN alike (0.54–0.66)
Members moderately active or not unusual (volume 4–8), dense rings	Shared patients	+0.36 to +0.61	GB-CAG or hybrids; pure GNNs 0.17–0.57 lower
Members not unusual (volume 4), sparse rings	Little	At most +0.03; all methods near chance	Hybrid slightly better (+0.032, confirmed)
Members not unusual (volume 4), claims mildly anomalous ($\delta = 1$), dense rings	Shared patients and claim attributes	+0.29 to +0.30	GB-CAG or hybrids; pure GNNs 0.11–0.29 lower
Members not unusual (volume 4), claims anomalous ($\delta \geq 1$), sparse rings	Claim attributes	At most +0.02	SAGE, HAN (and RGCN at $\delta = 2$) and logistic regression, +0.06 to +0.18 over GB-CAG
One claim per member (volume 2)	None detectable	None	No method
Other shapes and sizes (hub, distributed, small and large block)	Shared patients; member volume for small and distributed rings	+0.06 to +0.49 (n.s. for small and distributed rings at full volume)	GB-CAG or hybrids; hybrid better for the distributed ring; pure GNNs no better
Public provider-labelled data	Provider volume; no label assortativity through shared beneficiaries	None (GB-PG −0.007, 25 rotations)	Gradient boosting on provider aggregates
Public data, expert group-fraud flags (100 large providers)	Provider volume	None (GB-PG −0.001, 25 rotations)	Gradient boosting on provider aggregates

Table A1 Generator parameters (defaults in `research/synth/generator.py`).

Parameter	Value
Patients / doctors / hospitals / regions / specialties	6,000 / 300 / 40 / 5 / 8
Audit periods	4 × 6 months (train: 1–2, validation: 3, test: 4)
Legitimate claims per patient	Poisson(gamma rate; mean 3, shape 1.5) over 24 months
Share of legitimate claims with a regular doctor	0.85 (1–3 regular doctors per patient, same region)
Claim amount	log-normal: specialty mean + 0.15 (procedures − 1) + 0.8 inpatient, s.d. 0.6
Isolated fraud	3% of legitimate volume; feature shift 2.0
Ring size and shape	4 doctors, 12 recruited patients (utilisation-weighted), linked as a block; shape and size experiments: hub (4 by 12), distributed (8 by 24, two ring doctors per patient), small (2 by 6) and large (8 by 24) blocks
Ring volume	16 claims per month during activity; monthly burst days (ring-volume experiments: 8, 4 and 2; shape experiments: the same or half the per-member volume)
Rings per period	4 (emergent: new members each period; persistent: same 4 rings)
Feature shift δ	0, 1, 2 (development worlds: 0.5)
Structural density s	0.1, 0.5, 0.9 (development worlds: 0.5)
Worlds per regime	12 (seeds 1–12; the regimes of a seed share its legitimate claims, isolated fraud and ring members; secondary analyses use seeds 1–6); confirmatory test: seeds 7–18 at volume 4; development worlds: seeds 901, 902

References

[1] National Health Care Anti-Fraud Association. The Challenge of Health Care Fraud. Accessed 20 October 2026. Available from: https://www.nhcaa.org/tools-insights/about-health-care-fraud/the-challenge-of-health-care-fraud/.

[2] Schlichtkrull M, Kipf TN, Bloem P, van den Berg R, Titov I, Welling M. Modeling relational data with graph convolutional networks. In: The Semantic Web (ESWC 2018). Lecture Notes in Computer Science, vol. 10843. Springer; 2018. p. 593–607. https://doi.org/10.1007/978-3-319-93417-4_38.

[3] Wang X, Ji H, Shi C, Wang B, Ye Y, Cui P, et al. Heterogeneous graph attention network. In: Proceedings of the World Wide Web Conference (WWW); 2019. p. 2022–2032. https://doi.org/10.1145/3308558.3313562.

[4] Hu Z, Dong Y, Wang K, Sun Y. Heterogeneous graph transformer. In: Proceedings of the Web Conference 2020 (WWW); 2020. p. 2704–2710. https://doi.org/10.1145/3366423.3380027.

[5] Lv Q, Ding M, Liu Q, Chen Y, Feng W, He S, et al. Are we really making much progress? Revisiting, benchmarking and refining heterogeneous graph neural networks. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining; 2021. p. 1150–1160. https://doi.org/10.1145/3447548.3467350.

[6] Lu J, Lin K, Chen R, Lin M, Chen X, Lu P. Health insurance fraud detection by using an attributed heterogeneous information network with a hierarchical attention mechanism. BMC Medical Informatics and Decision Making. 2023;23:62. https://doi.org/10.1186/s12911-023-02152-0.

[7] Hong B, Lu P, Xu H, Lu J, Lin K, Yang F. Health insurance fraud detection based on multi-channel heterogeneous graph structure learning. Heliyon. 2024;10(9):e30045. https://doi.org/10.1016/j.heliyon.2024.e30045.

[8] Muhammad R, Tbaishat D, Nazir A, Yacoub S, AbdulRazek M, El-Enen MAA, et al. Fraud detection and explanation in medical claims using GNN architectures. Scientific Reports. 2025;15:41734. https://doi.org/10.1038/s41598-025-22910-6.

[9] Grinsztajn L, Oyallon E, Varoquaux G. Why do tree-based models still outperform deep learning on typical tabular data? In: Advances in Neural Information Processing Systems 35 (NeurIPS), Datasets and Benchmarks Track; 2022.

[10] Weber M, Domeniconi G, Chen J, Weidele DKI, Bellei C, Robinson T, et al. Anti-money laundering in Bitcoin: experimenting with graph convolutional networks for financial forensics. In: KDD Workshop on Anomaly Detection in Finance; 2019. arXiv:1908.02591. https://doi.org/10.48550/arXiv.1908.02591.

[11] Tang J, Hua F, Gao Z, Zhao P, Li J. GADBench: revisiting and benchmarking supervised graph anomaly detection. In: Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track; 2023.

[12] Yoo Y, Shin J, Kyeong S. Medicare fraud detection using graph analysis: a comparative study of machine learning and graph neural networks. IEEE Access. 2023;11:88278–88294. https://doi.org/10.1109/ACCESS.2023.3305962.

[13] Ma J, Li F, Zhang R, Xu Z, Cheng D, Ouyang Y, et al. Fighting against organized fraudsters using risk diffusion-based parallel graph neural network. In: Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence (IJCAI); 2023. p. 6138–6146. https://doi.org/10.24963/ijcai.2023/681.

[14] Curtis ED, Billion-Polak P, Khoshgoftaar TM, Furht B. A review of distinct machine learning classifiers for healthcare fraud detection. Journal of Big Data. 2025;12:238. https://doi.org/10.1186/s40537-025-01295-3.

[15] Kaufman S, Rosset S, Perlich C, Stitelman O. Leakage in data mining: formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data. 2012;6(4):15. https://doi.org/10.1145/2382577.2382579.

[16] Kapoor S, Narayanan A. Leakage and the reproducibility crisis in machine-learning-based science. Patterns. 2023;4(9):100804. https://doi.org/10.1016/j.patter.2023.100804.

[17] Bauder RA, Khoshgoftaar TM. Medicare fraud detection using machine learning methods. In: Proceedings of the 16th IEEE International Conference on Machine Learning and Applications (ICMLA); 2017. p. 858–865. https://doi.org/10.1109/ICMLA.2017.00-48.

[18] Herland M, Khoshgoftaar TM, Bauder RA. Big data fraud detection using multiple Medicare data sources. Journal of Big Data. 2018;5:29. https://doi.org/10.1186/s40537-018-0138-3.

[19] Johnson JM, Khoshgoftaar TM. Medicare fraud detection using neural networks. Journal of Big Data. 2019;6:63. https://doi.org/10.1186/s40537-019-0225-0.

[20] Thornton D, Mueller RM, Schoutsen P, van Hillegersberg J. Predicting healthcare fraud in Medicaid: a multidimensional data model and analysis techniques for fraud detection. Procedia Technology. 2013;9:1252–1264. https://doi.org/10.1016/j.protcy.2013.12.140.

[21] Akoglu L, Tong H, Koutra D. Graph based anomaly detection and description: a survey. Data Mining and Knowledge Discovery. 2015;29(3):626–688. https://doi.org/10.1007/s10618-014-0365-y.

[22] Pourhabibi T, Ong KL, Kam BH, Boo YL. Fraud detection: a systematic literature review of graph-based anomaly detection approaches. Decision Support Systems. 2020;133:113303. https://doi.org/10.1016/j.dss.2020.113303.

[23] Akoglu L, McGlohon M, Faloutsos C. OddBall: spotting anomalies in weighted graphs. In: Advances in Knowledge Discovery and Data Mining (PAKDD). Lecture Notes in Computer Science, vol. 6119. Springer; 2010. p. 410–421. https://doi.org/10.1007/978-3-642-13672-6_40.

[24] Beutel A, Xu W, Guruswami V, Palow C, Faloutsos C. CopyCatch: stopping group attacks by spotting lockstep behavior in social networks. In: Proceedings of the 22nd International Conference on World Wide Web (WWW); 2013. p. 119–130. https://doi.org/10.1145/2488388.2488400.

[25] Hooi B, Song HA, Beutel A, Shah N, Shin K, Faloutsos C. FRAUDAR: bounding graph fraud in the face of camouflage. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining; 2016. p. 895–904. https://doi.org/10.1145/2939672.2939747.

[26] Wang J, Wen R, Wu C, Huang Y, Xiong J. FdGars: fraudster detection via graph convolutional networks in online app review system. In: Companion Proceedings of the 2019 World Wide Web Conference; 2019. p. 310–316. https://doi.org/10.1145/3308560.3316586.

[27] Zhang Y, Fan Y, Ye Y, Zhao L, Shi C. Key player identification in underground forums over attributed heterogeneous information network embedding framework. In: Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM); 2019. p. 549–558. https://doi.org/10.1145/3357384.3357876.

[28] Dou Y, Liu Z, Sun L, Deng Y, Peng H, Yu PS. Enhancing graph neural network-based fraud detectors against camouflaged fraudsters. In: Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM); 2020. p. 315–324. https://doi.org/10.1145/3340531.3411903.

[29] Liu Y, Ao X, Qin Z, Chi J, Feng J, Yang H, et al. Pick and choose: a GNN-based imbalanced learning approach for fraud detection. In: Proceedings of the Web Conference 2021 (WWW); 2021. p. 3168–3177. https://doi.org/10.1145/3442381.3449989.

[30] Tang J, Li J, Gao Z, Li J. Rethinking graph neural networks for anomaly detection. In: Proceedings of the 39th International Conference on Machine Learning (ICML). PMLR, vol. 162; 2022. p. 21076–21089.

[31] Mardani S, Moradi H. Using graph attention networks in healthcare provider fraud detection. IEEE Access. 2024;12:132786–132800. https://doi.org/10.1109/ACCESS.2024.3425892.

[32] Mardani S, Moradi H, Asadpour M. Modeling interdependencies in fraud detection: a graph embedding-based approach. Complex & Intelligent Systems. 2026;12(9):220. https://doi.org/10.1007/s40747-026-02336-x.

[33] Zhou J, Wang X, Wang J, Ye H, Wang H, Zhou Z, et al. FraudAuditor: a visual analytics approach for collusive fraud in health insurance. IEEE Transactions on Visualization and Computer Graphics. 2023;29(6):2849–2861. https://doi.org/10.1109/TVCG.2023.3261910.

[34] Liu J, Bier E, Wilson A, Guerra-Gomez JA, Honda T, Sricharan K, et al. Graph analysis for detecting fraud, waste, and abuse in health-care data. AI Magazine. 2016;37(2):33–46. https://doi.org/10.1609/aimag.v37i2.2630.

[35] Bazhenov G, Platonov O, Prokhorenkova L. GraphLand: evaluating graph machine learning models on diverse industrial data. In: Advances in Neural Information Processing Systems 38 (NeurIPS), Datasets and Benchmarks Track; 2025. arXiv:2409.14500. https://doi.org/10.48550/arXiv.2409.14500.

[36] Vandervorst F, Deprez B, Verbeke W, Verdonck T. Inductive inference of gradient-boosted decision trees on graphs for insurance fraud detection. Data Mining and Knowledge Discovery. 2026;40(5):78. https://doi.org/10.1007/s10618-026-01249-0.

[37] Chen Z, Chen L, Villar S, Bruna J. Can graph neural networks count substructures? In: Advances in Neural Information Processing Systems 33 (NeurIPS); 2020. arXiv:2002.04025. https://doi.org/10.48550/arXiv.2002.04025.

[38] Altman E, Blanuša J, von Niederäusern L, Egressy B, Anghel A, Atasu K. Realistic synthetic financial transactions for anti-money laundering models. In: Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track; 2023.

[39] Jesus S, Pombal J, Alves D, Cruz A, Saleiro P, Ribeiro R, et al. Turning the tables: biased, imbalanced, dynamic tabular datasets for ML evaluation. In: Advances in Neural Information Processing Systems 35 (NeurIPS), Datasets and Benchmarks Track; 2022.

[40] Shchur O, Mumme M, Bojchevski A, Günnemann S. Pitfalls of graph neural network evaluation. In: NeurIPS Workshop on Relational Representation Learning; 2018. arXiv:1811.05868. https://doi.org/10.48550/arXiv.1811.05868.

[41] Platonov O, Kuznedelev D, Diskin M, Babenko A, Prokhorenkova L. A critical look at the evaluation of GNNs under heterophily: are we really making progress? In: International Conference on Learning Representations (ICLR); 2023.

[42] Xu K, Hu W, Leskovec J, Jegelka S. How powerful are graph neural networks? In: International Conference on Learning Representations (ICLR); 2019.

[43] Ke G, Meng Q, Finley T, Wang T, Chen W, Ma W, et al. LightGBM: a highly efficient gradient boosting decision tree. In: Advances in Neural Information Processing Systems 30 (NeurIPS); 2017. p. 3146–3154.

[44] Hamilton WL, Ying R, Leskovec J. Inductive representation learning on large graphs. In: Advances in Neural Information Processing Systems 30 (NeurIPS); 2017. p. 1024–1034.

[45] Pedregosa F, et al. Scikit-learn: machine learning in Python. Journal of Machine Learning Research. 2011;12:2825–2830.

[46] Fey M, Lenssen JE. Fast graph representation learning with PyTorch Geometric. In: ICLR Workshop on Representation Learning on Graphs and Manifolds; 2019. arXiv:1903.02428. https://doi.org/10.48550/arXiv.1903.02428.

[47] Saito T, Rehmsmeier M. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLoS ONE. 2015;10(3):e0118432. https://doi.org/10.1371/journal.pone.0118432.

[48] Dal Pozzolo A, Boracchi G, Caelen O, Alippi C, Bontempi G. Credit card fraud detection: a realistic modeling and a novel learning strategy. IEEE Transactions on Neural Networks and Learning Systems. 2018;29(8):3784–3797. https://doi.org/10.1109/TNNLS.2017.2736643.

[49] Holm S. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics. 1979;6(2):65–70.

[50] Demšar J. Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research. 2006;7:1–30.

[51] Gupta RA. Healthcare Provider Fraud Detection Analysis (dataset). Kaggle; 2019. CC0 1.0. Accessed 20 October 2026. Available from: https://www.kaggle.com/datasets/rohitrox/healthcare-provider-fraud-detection-analysis.

[52] Nadeau C, Bengio Y. Inference for the generalization error. Machine Learning. 2003;52(3):239–281. https://doi.org/10.1023/A:1024068626366.

[53] Bouckaert RR, Frank E. Evaluating the replicability of significance tests for comparing learning algorithms. In: Dai H, Srikant R, Zhang C, editors. Advances in Knowledge Discovery and Data Mining (PAKDD 2004). Lecture Notes in Computer Science. Springer; 2004. p. 3–12. https://doi.org/10.1007/978-3-540-24775-3_3.

[54] Rossi E, Chamberlain B, Frasca F, Eynard D, Monti F, Bronstein M. Temporal graph networks for deep learning on dynamic graphs. arXiv:2006.10637; 2020. https://doi.org/10.48550/arXiv.2006.10637.

[55] Walonoski J, Kramer M, Nichols J, Quina A, Moesel C, Hall D, et al. Synthea: an approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. Journal of the American Medical Informatics Association. 2018;25(3):230–238. https://doi.org/10.1093/jamia/ocx079.

[56] Liu K, Dou Y, Zhao Y, Ding X, Hu X, Zhang R, et al. BOND: benchmarking unsupervised outlier node detection on static attributed graphs. In: Advances in Neural Information Processing Systems 35 (NeurIPS), Datasets and Benchmarks Track; 2022. arXiv:2206.10071. https://doi.org/10.48550/arXiv.2206.10071.

[57] Kapoor S, Cantrell EM, Peng K, et al. REFORMS: consensus-based recommendations for machine-learning-based science. Science Advances. 2024;10(18):eadk3452. https://doi.org/10.1126/sciadv.adk3452.

[58] Gebru T, Morgenstern J, Vecchione B, Vaughan JW, Wallach H, Daumé III H, et al. Datasheets for datasets. Communications of the ACM. 2021;64(12):86–92. https://doi.org/10.1145/3458723.